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Aria-NBV: Quality-Driven Next-Best-View Planning with Egocentric Foundation Models

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Transparency in the Use of AI Tools

AI-assisted tools were used to organize literature notes, check consistency across repository documentation, and draft parts of the thesis seed. The author remains responsible for the final research scope, technical claims, citations, implementation, experiments, and submitted document.

Validation TODO: Update this statement against final institutional requirements before submission.

Source: proposal transparency text

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Abstract

This thesis investigates target-conditioned, quality-driven Next-Best-View (NBV) planning for egocentric 3D reconstruction in Aria Synthetic Environments (ASE). It uses target-specific Relative Reconstruction Improvement (RRI) as the supervision and evaluation signal for finite candidate view selection, measures bounded oracle-lookahead headroom, and tests whether an actor-visible finite-horizon value model can recover part of that headroom under matched oracle re-evaluation.

Zusammenfassung

Open question: Write the German abstract after the English thesis claim and final evidence scale are stable.

Source: main thesis seed

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1 Introduction

Active reconstruction is a coupled sensing and inference problem: the reconstruction error after a fixed acquisition budget depends as much on the selected viewpoints as on the downstream surface estimator. Classical active perception treats sensing actions as part of perception itself, and view-planning surveys formalize the generate-score-select loop for three-dimensional inspection [AWB88, Baj88, SRR03]. ARIA-NBV adopts that loop for egocentric indoor data, but asks a narrower question: when views are restricted to a finite feasible candidate table, does reconstruction-quality improvement contain planning structure beyond one-step selection?

The key empirical precedent is VIN-NBV, which replaces pure coverage with RRI, an oracle label computed from point-mesh reconstruction-error reduction after adding a query view [Fra+25]. ARIA-NBV keeps this quality-driven axis because target indoor surfaces can remain poor even when coverage proxies look saturated. The thesis transfers this idea to the Project Aria and ASE regime, where calibrated egocentric streams, trajectories, semidense points, predicted object boxes, and mesh-supervised assets support controlled oracle labels [Eng+23, Met25a, Str+24a].

The implemented seminar substrate already provides scene-level oracle RRI labels and a VINv3-style one-step candidate scorer on frozen EVL features. This thesis treats that substrate as a starting point, not the final result. The extension is target-specific: oracle target-task sampling for data generation, target-cropped labels, replayable counterfactual rollouts, and a finite-horizon value model over candidate rows. V1 actor-visible claims are evaluated through observed or predicted target descriptors and Ground-Truth Target Evaluation (GT-EVAL), not by giving GT targets to the actor.

Continuous and hierarchical NBV papers motivate later directions, but they do not define the first thesis test. GenNBV motivates continuous policy pressure, while Hestia motivates a target-then-pose hierarchy after the finite-candidate protocol is stable [Che+24, Lu+26]. Active NeRF, FisherRF, Next Best Sense,

semantic best-view, object-centric NBV, and FOVHPE motivate uncertainty, information, semantic, and target-aware diagnostics, but those channels are compared against the mesh-supervised RRI objective rather than replacing it [BLL25, Jeo+26, JLD24, LJD25, Pan+22, Str+24b].

THESIS POSITION

ARIA-NBV tests target-conditioned, quality-driven NBV on ASE/EFM3D as a finite-candidate planning problem. The core experiment first measures whether bounded oracle lookahead exposes non-myopic target-RRI headroom. If it does, a masked candidate-query Q_H model is evaluated by how much of that headroom it recovers from actor-visible rollout data.

The system object is a leakage-safe finite-candidate target-specific RRI decision process on ASE/EVL. Given an ASE snippet, an oracle-sampled target task, a finite valid candidate set, and a fixed horizon, ARIA-NBV measures target-specific point-mesh RRI, estimates non-myopic headroom by oracle lookahead, and learns Q_H from non-privileged state and target-task descriptors to recover that headroom. The learned state-action value model induces the finite-action policy π_Q .

The contribution is fourfold: a leakage-safe target-specific RRI protocol, a headroom-measured finite-candidate planning test, a geometry-aware residual Q_H value model, and support-aware rollout generation. The validation sequence follows that order: target-specific RRI label construction, a calibrated target-conditioned one-step scorer over potential counterfactual views, oracle-lookahead headroom, and residual dueling Q_H over finite valid candidates.

Open decision: The thesis title, exact RQ wording, and final evidence scale must be re-locked once M5 evidence is known.

Source: proposal sections; advisor handout

Gate: pre-submission thesis freeze

2 Background

The project combines ASE scene assets, Egocentric Foundation Model 3D (EFM3D) features, oracle RRI labels, and VIN-style candidate scoring. ASE supplies Aria-like synthetic sensor trajectories, aligned semi-dense maps, and GT annotations at indoor-scene scale [Met25a]. EVL supplies frozen local voxel evidence, lifted DINO-derived features, and object-support signals for target-task descriptors and diagnostics [Met25b, Str+24a]. In the current thesis scope, EVL is local target/support evidence rather than a full long-horizon scene memory; broad scene context comes from semi-dense and fused geometry, with point-attached feature banks treated as planned representation ablations until a persisted cache and training reader exist.

2.1 Related Work

ARIA-NBV sits between classical view planning, learned reconstruction-quality scoring, and egocentric scene understanding. Classical NBV work formalizes the generate-score-select loop for three-dimensional inspection and reconstruction, while submodular and adaptive-submodular active-sensing results explain why greedy policies can be strong when the utility has diminishing returns [GK11, KSG08, SRR03]. This thesis uses that lineage as a baseline discipline rather than as a theorem claim: target-specific RRI depends on masks, object matching, candidate regeneration, occlusion, and changing support, so coverage-style approximation guarantees are treated as motivation for greedy controls, not as properties of the ARIA objective.

The closest objective precedent is VIN-NBV, which learns to rank sampled candidate views by oracle RRI instead of by pure coverage [Fra+25]. ARIA-NBV keeps that reconstruction-quality signal but changes both the state and the unit of evaluation. The actor-visible state comes from Project Aria-style egocentric streams and ASE assets [Eng+23, Met25a], and the thesis target is Target-Specific RRI (target RRI) rather than only scene-level gain. CORAL-style ordinal learning

is therefore useful for a calibrated one-step scorer, but it is a control for label reliability and ranking, not the whole finite-horizon planning result [CMR19].

Project Aria, ASE, and EFM3D supply the substrate for this thesis. Project Aria defines the multimodal egocentric sensor regime and calibration assumptions, ASE provides mesh-supervised synthetic trajectories and annotations, and EFM3D/Egocentric Voxel Lifting (EVL) contributes actor-visible local voxel evidence, lifted image features, and object hypotheses [Eng+23, Met25a, Str+24a]. The thesis does not treat EVL as a complete world model. Its role is local target/support evidence; broader scene memory is represented through semi-dense or fused point support and, if needed, separate point-attached feature banks.

Coverage and information-based NBV methods provide useful diagnostic channels but not the primary reward. SCONE and MACARONS estimate surface visibility or coverage from learned scene representations, while FisherRF and Next Best Sense use information gain and uncertainty in radiance-field or Gaussian-splatting settings [GML22, Gué+23, JLD24, Str+24b]. ARIA-NBV can adopt support, overlap, uncertainty, and directional-history features from this family, but the thesis compares those proxies against target RRI rather than replacing the mesh-supervised label. Object-centric and semantic Gaussian-splatting work similarly motivates target-aware utility and semantic failure analysis, while remaining a bridge once the mesh/oracle target protocol is stable [Jeo+26, Ker+23].

Continuous-policy NBV papers define an important boundary for the thesis. GenNBV learns a generalizable continuous policy for active reconstruction, PB-NBV emphasizes efficient projection-based candidate scoring, and Hestia introduces a hierarchical target-then-pose formulation with directional observability [Che+24, Jia+25, Lu+26]. These works motivate later actor-critic, projection-shortlist, and hierarchy experiments. The core thesis remains narrower: finite candidate tables over ASE snippets, oracle target labels from GT-EVAL, and a masked Q_H model whose selected actions are re-evaluated by the same oracle.

The planned value model also borrows structure from set and geometric learning. Deep Sets gives the minimum symmetry contract for unordered candidate

or point sets; Set Transformer supplies masked candidate-candidate interaction; and query-centric trajectory forecasting motivates local relative positional encodings without importing a motion-forecasting decoder [Lee+19, Zah+17, Zho+23]. Geometric deep learning frames the broader design question as matching model symmetries to the output map, not making the whole system invariant [Bro+21]. For ARIA-NBV this means candidate-row permutation equivariance for Q_H scores, mask isolation for invalid rows, explicit target/current/candidate frames, and gravity-aware rather than full rotation-invariant features. The dedicated theory treatment in Section 2.2 turns those literature roles into the architecture ladder and acceptance tests.

Point and sparse scene encoders are best treated as representation ablations after the simpler support contract is measured. Point Transformer, PointNeXt, Point Transformer V3, KPConv, and sparse-convolutional encoders offer local geometric interaction, density handling, scalability, or sparse-volume computation [CGS19, Qia+22, Tho+19, Wu+24, Zha+21]. They become relevant if compact target, frustum, support, and lifted-feature pools leave a measurable bottleneck for target RRI ranking or finite-horizon Q_H . They are not first-line replacements for EFM3D because a thesis backbone must still provide actor-visible OBB hypotheses and reusable scene conditioning.

Finally, offline and recurrent decision-modeling papers inform the training and ablation ladder without changing the problem definition. Double DQN and dueling heads motivate selector/evaluator separation and value/advantage structure for finite candidate tables [HGS15, Wan+16]. IQL, CQL, and BCQ motivate explicit support constraints and skepticism about out-of-distribution bootstrapping, not an immediate switch away from the finite-candidate replay contract [FMP19, KNL21, Kum+20]. Trajectory Transformer, Decision Transformer, and Gumbel-Top-k provide vocabulary for sequence modeling, return conditioning, and stochastic branch generation only after the target/candidate/mask/evaluation contract is shared with the simpler baselines [Che+21, JLL21, KHW19]. Deja View suggests a bounded weight-tied refinement ablation for candidate-context

tokens, but not a claim that reconstruction-loop recurrence transfers directly to ARIA-NBV planning [Bur+26].

Literature family	Useful contribution	ARIA-NBV boundary
VIN-NBV and ordinal scoring	Oracle RRI labels, one-step candidate ranking, and CORAL-style calibration.	Adopt as target-specific label/scorer controls; do not stop at myopic ranking if oracle lookahead exposes headroom.
Aria, ASE, EFM3D/EVL	Egocentric streams, calibration, semi-dense support, local voxel evidence, and OBB hypotheses.	Use as actor-visible substrate; keep GT meshes and target crops for labels/evaluation only.
SCONE, MAC-ARONS, FisherRF, 3DGS NBV	Coverage, support, uncertainty, information gain, and directional novelty channels.	Use as diagnostics or candidate-token features; do not replace target RRI as the thesis objective.
Set/geometric models	Permutation structure, masked attention, query-local relative encodings, and typed symmetry tests.	Use for finite-candidate Q_H controls before point/sparse back-bone escalation.
Continuous and recurrent bridges	GenNBV/Hestia continuous or hierarchical policies; Deja View-style refinement.	Defer to bridge/ablation work after finite-candidate labels, masks, and replay are trustworthy.

Table 1: Adopted roles and explicit boundaries for the literature used by the thesis.

2.2 Geometric Learning and Candidate-Set Theory

The geometric-learning question in ARIA-NBV is not whether a more expressive scene backbone can be added to the system. The thesis object is a typed finite-candidate decision map: for a target task, selected history, partial geometry, and an unordered set of feasible candidate poses, predict which candidate will improve target reconstruction under oracle re-evaluation. The model should therefore match the symmetries of this map before it escalates to heavier point, sparse, recurrent, or equivariant modules [Bro+21].

The first required structure is candidate-row permutation equivariance. Reordering the rows of Q_t may reorder per-candidate Q_H outputs, but it must not change the value assigned to the same physical candidate. Deep Sets supplies

the minimal invariant/equivariant baseline for unordered candidate and support sets, while Set Transformer supplies masked candidate-candidate interaction once independent and pooled controls are calibrated [Lee+19, Zah+17]. The thesis should not replace the row path with only a pooled scene embedding, because the policy selects a candidate row.

The minimum algebraic contract is therefore a row-wise map over a masked set. For any permutation matrix Π acting on candidate rows, the value model should satisfy

$$f_{\theta}(\Pi \mathbf{X}_t, \Pi \mathbf{m}_t) = \Pi f_{\theta}(\mathbf{X}_t, \mathbf{m}_t),$$

where $\mathbf{X}_t = \{\mathbf{x}_{t,i}\}_{i=1}^{N_q}$ stores per-candidate actor-visible descriptors and $\mathbf{m}_t$ stores validity. Invalid rows are not low-quality actions; they are outside the admissible set. A masked scorer must therefore compute selection over

$$\mathcal{A}_t = \{i \in \{1, \dots, N_q\} : m_{t,i} = 1\}, \quad a_t^{\theta} = \operatorname{argmax}_{i \in \mathcal{A}_t} f_{\theta,i}(\mathbf{X}_t, \mathbf{m}_t).$$

The local-frame contract is a controlled gauge choice rather than a full invariance claim. Candidate features may use transforms such as $\mathbf{T}_{c_q}^r$, target-relative bearings, and query-local relative encodings so that a global origin change does not change the physical row identity, while gravity alignment and camera-frustum geometry remain available task structure [Zho+19, Zho+23].

The second required structure is gauge discipline. Candidate, target, current camera, and selected-history poses should be encoded in local frames, with continuous rotation representations and query-local relative positional features where needed [Zho+19, Zho+23]. This borrows the useful part of query-centric trajectory models: local relations and relative encodings. It does not import their forecasting decoder, road-agent priors, or online streaming claims. Full global SE(3) equivariance is also not the right default because the task is gravity aligned, camera-frustum limited, and tied to egocentric acquisition. Exact equivariant networks remain ablations for support encoders or candidate graphs, not the core claim [Fuc+20, SHW21].

The third required structure is directional visibility memory. A candidate view is not only a posed pinhole pose; it is a directional field from which target-local surface regions may or may not have been observed. This suggests storing selected-view history on $\mathbb{S}^2$ through a histogram, a second-moment summary, or low-order spherical harmonics before more specialized equivariant layers are introduced [e3n25]. Coverage and information-gain methods motivate support, overlap, and uncertainty features, but those features remain diagnostic channels unless they improve target-specific oracle RRI under the paired evaluation protocol [GML22, JLD24].

Object	Required structure	Testable consequence
Candidate rows	Permutation-equivariant per-row value map.	Row-shuffle tests must satisfy $f_{\theta(\Pi X, \Pi m)} = \Pi f_{\theta(X, m)}$ for every selection score.
Invalid and padded rows	Hard mask isolation.	Invalid rows cannot change valid-row scores except through explicit valid-count or support features.
Candidate-target geometry	Target-local and candidate-local relative pose features.	World-frame origin, yaw convention, and display-only camera transforms cannot become shortcuts.
Selected-view history	Directional memory on $\mathbb{S}^2$.	Novelty tests separate already-observed target directions from generic pose distance.
Actor/oracle boundary	Typed provenance for target descriptors, labels, and support features.	GT meshes, crops, and all-candidate renders supervise labels/evaluation only.

Table 2: Minimum geometric-learning contract for the finite-candidate value model.

This contract implies an architecture ladder with conservative attribution. The independent scorer tests whether candidate self-features already solve the target task. A pooled DeepSets context tests whether valid-set summaries help without pairwise attention. A masked Set Transformer tests candidate-candidate interaction. QCNet-style relative positional encoding tests whether local candidate-target/history relations are the missing signal. Directional support and overlap biases test whether visibility memory or surface-support hypotheses add value.

Only after these controls pass should point backbones, sparse convolutions, exact equivariant graphs, or looped refinement be treated as thesis-relevant architecture changes [Bur+26, CGS19, Qia+22, Tho+19, Wu+24].

Level	Model	Scientific role
A0	Independent candidate scorer	Locks target/candidate descriptors, CORAL calibration, and one-step ranking before context is added.
A1	Pooled DeepSets context	Tests unordered valid-set summaries while preserving a per-candidate row path.
A2	Masked Set Transformer	Tests candidate-candidate interaction under row-shuffle and mask-isolation checks.
A3	Query-local relative encoding	Adapts QCNet-style local RPE for candidate-target, candidate-candidate, and selected-history relations.
A4	Directional/support bias	Tests visibility memory, target-frustum support, overlap, and uncertainty as features rather than proxy rewards.
A5	Residual dueling Q_H	Tests finite-horizon recovery over the calibrated myopic scorer with masked Double-Q backups.
A6+	Point/sparse/exact-equivariant/recurrent encoders	Escalates only if compact descriptors bottleneck target-RRI ranking or Q_H recovery.
B0	Online candidate-table refresh	Bridge level after offline Q_H evidence: tests whether the same finite-action contract survives streaming candidate regeneration.
B1	Continuous policy / simulator controller	Post-thesis bridge only unless finite candidates fail; requires a separate action-parameterization, safety, and simulator-realism gate.

Table 3: Architecture ladder used to keep geometric-learning claims attributable.

The active method turns this theory into concrete replay fields, descriptor versions, and acceptance tests in Chapter Section 3.5. The literature review uses the same split: Deep Sets, Set Transformer, and query-local relative encoding are immediate model controls; point, sparse, exact-equivariant, radiance-field, and recurrent

models are scoped ablations or bridge work until the finite-candidate target-RRI protocol is stable.

Validation TODO: Add measured row-shuffle, mask-isolation, duplicate-row, frame-transform, and valid-count stress-test results before claiming any geometric-learning module improves Q_H .

Source: autoresearch thesis-lit-review report; docs/contents/theory/candidate_view_dependence.qmd

Gate: architecture ablation evidence

The literature is used here to assign roles, not to broaden the thesis claim. Older active perception motivates action-conditioned sensing; VIN-NBV supplies the quality-driven candidate-ranking precedent; Project Aria, ASE, and EFM3D motivate the logged egocentric state; and offline value learning supplies replay and overestimation controls for the finite candidate table.

Role		Relevant signal	Adopt / defer
Quality-driven	NBV	Oracle RRI and ordinal one-	Adopt point-mesh RRI labels and
	[Fra+25]	step candidate ranking are	a learned one-step target scorer;
		the closest implemented prece-	test whether one-step ranking is
		dent.	enough.
Egocentric	substrate	Logged streams and tra-	Use observed/predicted target de-
	[Eng+23,	jectory, semi-dense support,	scriptors as actor input; keep GT
	Met25a,	frozen EVL/EFM3D evidence,	geometry and crops for labels and
	Str+24a]	and observed/predicted OBBs	evaluation.
		form the actor-visible state.	
Greedy sensing and fi-		Diminishing-returns intuition;	Measure oracle-lookahead head-
nite candidates	[GK11,	oracle-lookahead headroom as	room before claiming a learnable
	KSG08]	a required test.	non-myopic advantage.
CORAL, set mod-		Ordinal target scorer, MLP/	QCNet trajectory decoding, mo-
els, QCNet, Double-		DeepSets/Set Transformer	tion-forecasting losses, online
Q	[CMR19,	candidate controls, query-cen-	streaming claims, IQL/CQL/
	HGS15,	tric relative-encoding ablation,	BCQ, and distributional heads re-
	Lee+19, Zho+23]	masked fitted backups.	main outside the initial value re-
			sult.
Continuous and ra-		Continuous policies, target-	Use as follow-up design pressure;
diance-field	NBV	then-pose hierarchies, target/	do not replace target RRI with
	[Jeo+26, Lu+26]	object-focused utility, and	coverage, uncertainty, or semantic
		uncertainty/semantic utility	proxy rewards.
		channels are useful compar-	
		isons.	

Table 4: Source-backed literature roles for the thesis scope.

The resulting lineage is deliberately narrow:

$$\mathcal{U}_{\text{cov/unc}} \rightarrow \hat{r}_t^e(i) \rightarrow r_t^e \rightarrow G_t^{(H)} \rightarrow Q_{H,\theta}.$$

Coverage and uncertainty remain diagnostics, not the thesis utility. GT meshes and GT target boxes remain oracle assets for target-task sampling, labels, and evaluation; they are not learned actor inputs unless a named privileged ablation says so. Offline and continuous RL references become meaningful only after candidate support, masks, and oracle re-evaluation are trustworthy.



Figure 1: Evidence chain from actor-visible state and target descriptor to masked candidates, target RRI, lookahead headroom, and the Q_H model. Dashed paths are follow-up work.

Hestia informs a deferred hierarchy in which a target or look-at point is proposed before choosing a feasible pose conditioned on it [Lu+26]. In ARIA-NBV this factorization keeps target-specific RRI as supervision/evaluation and treats feasibility projection or masks as constraints:

$$\pi_{\text{cont}}(\mathbf{x}_t, \ell_t \mid s_t, z_e) = \pi_{\text{pose}}(\mathbf{x}_t \mid \ell_t, s_t, z_e) \pi_{\text{look}}(\ell_t \mid s_t, z_e)$$

Research TODO: Decide which bridge literature belongs in the final thesis background versus the discussion/future-work chapter.

Source: proposal related work; advisor handout

Gate: final literature pass

3 Method

ARIA-NBV is a target-conditioned, finite-candidate NBV problem in the Project Aria / ASE / EFM3D observation regime. The thesis is deliberately not a first-order continuous-control claim: it tests whether bounded planning over a finite valid candidate table improves target reconstruction quality beyond myopic view selection.

3.1 Formal State and Visibility

The initial non-myopic experiment is a *masked finite-horizon candidate-decision process*, not a stationary deployment MDP. It is an offline, mesh-supervised *controlled counterfactual replay* process: the oracle may render selected candidates and regenerate the next candidate table as part of the experimental protocol, while the actor sees only the typed state features below.

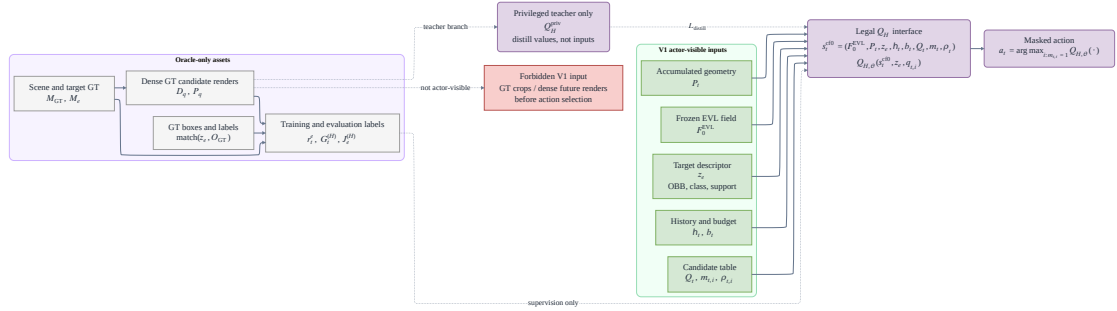


Figure 2: Actor-visible and oracle-only state boundary for the V1 thesis protocol. The learned Q_H policy may consume accumulated geometry, frozen EVL evidence, target descriptors, history, budget, candidates, masks, and invalid reasons; GT meshes, boxes, dense candidate renders, target labels, and endpoint metrics supervise labels, upper bounds, evaluation, or explicitly named teacher ablations only.

The three state spaces referenced by the protocol tuple are the codomains of the shared state equations. The logged historic state contains only the original trajectory evidence,

$$s_t^{\text{hist}} = (\mathbf{I}^{\text{rgb}}, \mathbf{X}, \mathcal{P}^{\text{semi}}, \mathbf{F}_v, e_t, b_t) \in \mathcal{S}^{\text{hist}}$$

the counterfactual actor state is the deployable state for Q_H decisions,

$$s_t^{\text{cf0}} = (\mathbf{F}_v^{\text{root}}, \mathcal{P}_t, \mathcal{Q}_t, m_{t,i}, \rho_{t,i}, e_t, b_t) \in \mathcal{S}^{\text{cf0}}$$

and the privileged oracle state is a label/evaluation state outside the actor input graph:

$$s_t^{\text{oracle}} = (s_t^{\text{cf+}}, \mathcal{M}^{\text{GT}}, \mathcal{M}_e^{\text{GT}}, \mathbf{D}_q, \mathcal{P}_q, \text{RRI}) \in \mathcal{S}^{\text{oracle}}$$

$$\mathcal{M}_{\text{NBV}} = (\mathcal{S}^{\text{hist}}, \mathcal{S}^{\text{cf0}}, \mathcal{S}^{\text{oracle}}, \{\mathcal{A}_t\}, T, r_t^e, \gamma, H).$$

The model separates logged snippet state, counterfactual actor state, and privileged oracle state because real egocentric trajectories contain modalities that are not available after synthetic view choices. The raw historic state is available on the logged ASE trajectory. The planner state keeps local root EVL evidence fixed while updating the fused geometry proxy, optional point-feature bank, selected-view history, remaining horizon metadata, target descriptor, candidates, masks, and reason codes. The privileged oracle state augments this with GT geometry, the matched target mesh, all-candidate rendered points, and oracle labels for target-task selection, label generation, upper-bound planning, and evaluation.

GT meshes, GT OBBs, GT crops, GT semantic labels, and all-candidate GT renders are oracle assets. They may define target tasks and labels, but they are not learned actor inputs unless a named upper-bound or privileged-teacher ablation explicitly says so. Invalidity is modeled as a constraint: masks apply before argmax, temperature softmax, loss targets, and bootstrap maximization. True infeasibility or absent evaluation samples are invalid rows; low immediate target support is reported as a diagnostic unless it prevents valid oracle evaluation.

3.2 Data Generation and Target-Specific RRI Labels

Target selection is part of the oracle data-generation pipeline. It samples supervised target tasks for target-conditioned NBV, rather than making a deployable claim that the learned actor discovers the target of interest on its own. The learned model receives a target descriptor and is evaluated by whether its selected views improve the matched target under oracle re-evaluation.

The target descriptor records the task and the cheap evidence available for diagnostics or later descriptor ablations:

$$\mathbf{z}_e = \varphi(\hat{\mathbf{B}}_e, \hat{\mathbf{y}}_e, \hat{p}_e, A_e^{\text{proj}}, n_e^{\text{semi}}, n_e^{\text{EVL}}, \mathbf{T}_e^{\text{rel}})$$

In this descriptor, $\hat{\mathbf{B}}_e$ is observed, predicted, or otherwise proposed OBB geometry for the target task; $\hat{\mathbf{y}}_e$ is class probabilities or class embedding; $\hat{p}_e$ is confidence; A_e^{proj} is projected area; n_e^{semi} and n_e^{EVL} are semidense and EVL support counts; and $\mathbf{T}_e^{\text{rel}}$ is relative target pose. These fields are not all target-task gates in the first implementation. Class, confidence, current projection, semidense support, EVL support, distance, and target bearing are retained as descriptor and audit fields so that later subsets can ask whether the target-conditioned model depends on semantic correctness or observation quality.

3.2.1 Seminar Oracle Substrate and Thesis Delta

The seminar paper is implemented evidence for the one-step scene-level oracle substrate, not the current thesis objective. Its labeler constructs a candidate table for an ASE snippet, renders candidate depth from the GT mesh with the PyTorch3D camera path, backprojects and fuses candidate points with the current semi-dense reconstruction, and computes point-mesh accuracy/completeness as RRI labels [Fra+25, Met25c]. ARIA-NBV reuses that substrate for label provenance, but changes the task unit: the thesis sampler first creates target tasks, target crops define the error surface, and selected counterfactual transitions are written to a standalone rollout store for finite-horizon Q_H .

Seminar material	Thesis adaptation	Placement
GT-mesh depth rendering and point-mesh RRI	Reused as label provenance; target crops replace whole-scene error as the primary objective.	Main data-generation method plus appendix camera-convention details.
Legacy shell candidate sampling	Kept as a historical/free-shell abstraction; current target-conditioned mixtures own the main candidate distribution.	Candidate/replay method and appendix, not a default-policy claim.
CORAL ordinal scorer and bin diagnostics	Reused for myopic target-scorer calibration and expected-RRI controls before residual Q_H .	Value-model method and evaluation evidence gates.
Immutable one-step VIN offline store	Kept separate from selected-transition rollouts.zarr; all-candidate labels train the scorer, selected transitions train bootstrapped Q_H .	Data-flow appendix and replay evidence matrix.
Run-specific W&B and cache-size notes	Historical diagnostics only until regenerated from current manifests and configs.	Appendix/open-work TODO, not final experiment evidence.

Table 5: How seminar-paper implementation evidence is adapted without drifting into thesis claims.

Conflict: Do not copy seminar scene-level RRI, legacy shell sampler, one-step VIN store, or run-specific CORAL/W&B results as current target-conditioned finite-horizon evidence. They are implementation substrate or historical diagnostics unless regenerated under the target-task and rollout-store protocol.

Source: docs/typst/seminar_paper/main.typ; docs/typst/seminar_paper/sections/05-oracle-rri.typ; docs/typst/seminar_paper/sections/07-training-objective.typ; docs/typst/seminar_paper/sections/12h-appendix-offline-cache.typ

Gate: final data-generation appendix and experiment-manifest refresh

3.2.2 Target Selection

Automatic target selection constitutes the first procedural layer in target-centric oracle data generation. Given a historic snippet and its egocentric encodings, the sampler chooses target tasks for which supervised target-conditioned NBV is meaningful. A useful target task must be identifiable, evaluable, and potentially action-sensitive: the oracle must know which GT object the task refers to, target-specific error must be computable, and at least some feasible candidate views

should expose non-marginal target-specific RRI after oracle evaluation. Near-solved targets are therefore not discarded before storage, but their low headroom must be measured and preserved as evidence.

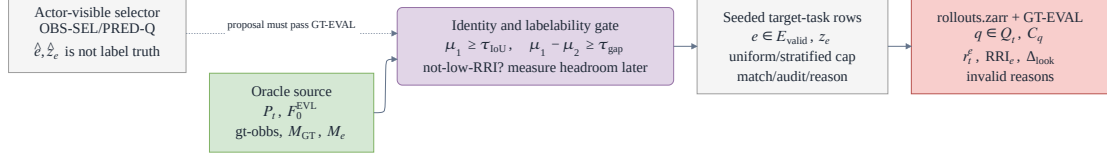


Figure 3: Oracle target-task sampling contract. GT OBBs and meshes define identity-valid supervised target tasks through an IoU and ambiguity-gap gate, a deterministic capped sampler writes target-task rows with descriptor and audit fields, and rollout generation later measures target RRI and headroom. The actor-visible target selector remains a separate diagnostic or later deployment contract, not the source of thesis labels.

The cheap admission gate is identity matching. A proposed target OBB is identity-valid when it matches exactly one GT target OBB by configured 3D IoU and ambiguity margin:

$$\mu_{\text{id}}(\hat{e}, e) = \text{IoU}_{3\text{D}}(\hat{\mathbf{B}}_{\hat{e}}, \mathbf{B}_e)$$

$$a_{\text{id}}(\hat{e}) = 1 \text{ iff } \begin{cases} \mu_1 \geq \tau_{\text{IoU}} \\ \mu_1 - \mu_2 \geq \tau_{\text{gap}} \end{cases}$$

Here μ_1 and μ_2 are the best and second-best target-to-GT IoU scores for the proposal. The threshold values are protocol parameters, not theory constants. The implementation should use a moderate default and report a threshold sweep so that coverage can be inspected under looser and stricter identity definitions.

Class prediction and confidence do not decide target-task eligibility in the first pass. They are recorded because semantic correctness may matter for target descriptors and failure analysis, but the oracle can compute target-specific RRI for a geometrically identified target even when the class head is noisy. Current RGB projection, semidense support, and EVL support are likewise audit fields rather than hard gates unless a later subset explicitly asks for projected-only or support-qualified targets.

The sampler keeps rollout cost bounded by selecting a capped number of identity-valid targets per source snippet. Within that cap, it samples uniformly with a deterministic seed rather than always taking the largest or highest-IoU object. The audit surface must still report how many identity-valid targets existed before the cap, how many were selected, and how their IoU, ambiguity gap, class, confidence, projected area, support, distance, and bearing were distributed.

Headroom is evaluated after target selection because it depends on oracle candidate scoring. The rollout store should persist identity-valid targets even when measured target headroom is low; training and evaluation loaders can then filter or stratify by headroom band. This preserves negative evidence: if many well-identified targets have no candidate with useful gain, the result is a candidate/support/headroom limitation rather than a target-selection failure hidden by pre-filtering.

Counterfactual trajectory naturalness is a candidate and rollout diagnostic, not an identity gate. Hard turns, target-bearing changes, and support collapse should be measured with candidate provenance and invalid-reason fields so that the thesis can distinguish target identity failures from unrealistic or unsupported view proposals.

3.2.3 Candidate View Generation

Candidate view generation is the second procedural layer: it turns one target task into a finite action table for one rollout state. The current thesis profile deliberately uses a small family mixture rather than the older unconstrained shell from the seminar paper. The data-generation config `.configs/build_rollouts_v1_realistic.toml` consumes the strict VIN offline store from `vin_offline`, uses the `train` split for the first real audit subset, samples one oracle target task per source sample, and writes a separate `rollouts_v1_realistic.zarr` store. The checked-in profile is an audit-scale thesis default with 25 source samples, not an LRZ path template and not final scale evidence.

At rollout step t , candidate generation constructs a full shell

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad N_q = 60,$$

with one fixed provenance component $k(i)$ per row. The canonical `v1_realistic_3family` mixture is

Component	Rows	Center family	View family
<code>forward_local</code>	24	<code>forward_local</code>	<code>forward_rig</code>
<code>target_bearing_local</code>	24	<code>target_bearing_local</code>	<code>target_point</code>
<code>lateral_target_bypass</code>	12	<code>lateral_target_bypass</code>	<code>target_point</code>

Table 6: Canonical three-family finite candidate table for real thesis rollouts. Counts are full-shell rows; valid-action counts are measured after geometry and motion pruning.

For each row, the raw direction is sampled in the reference rig frame. The realistic profile uses the forward-biased Power Spherical distribution from the current implementation [DAT20]:

$$\mathbf{u}_i \sim \text{PS}(\mathbf{e}_z, \kappa), \quad p(\mathbf{u}) = c_\kappa (1 + \mathbf{e}_z^T \mathbf{u})^\kappa, \quad \kappa = 8.$$

The draw is mapped into configured azimuth and elevation caps without rejection. With $\psi = \text{atan2}(u_x, u_z)$ and $u_y = \sin \theta$, the cap transform is

$$\psi' = \psi \frac{\Delta\psi}{2\pi}, \quad y' = \sin \theta_{\min} + \frac{u_y + 1}{2} \cdot (\sin \theta_{\max} - \sin \theta_{\min}),$$

$$\mathbf{d}_i^0 = \left\| \left(\sqrt{1 - y'^2} \sin \psi', y', \sqrt{1 - y'^2} \cos \psi' \right) \right\|.$$

The three position families then reinterpret this capped direction. Let $\mathbf{f} = \mathbf{e}_z$ be the rig-forward unit vector, $\mathbf{b}_e$ the actor-visible target bearing in the reference frame, $\mathbf{l}_e = \|\mathbf{e}_y \times \mathbf{b}_e\|$ the horizontal lateral direction, and $\mathbf{e}_y$ the world-up direction expressed in the sampling frame. The family directions are:

$$\mathbf{d}_i^{\text{forward}} = \|\mathbf{f} + \alpha_f (\mathbf{d}_i^0 - (\mathbf{d}_i^0 \cdot \mathbf{f}) \mathbf{f})\|, \quad \alpha_f = 0.45,$$

$$\mathbf{d}_i^{\text{target}} = \|\mathbf{b}_e + \alpha_t (\mathbf{d}_i^0 - (\mathbf{d}_i^0 \cdot \mathbf{b}_e) \mathbf{b}_e)\|, \quad \alpha_t = 0.4,$$

$$\mathbf{d}_i^{\text{bypass}} = \left\| 0.55 \mathbf{b}_e + 0.85 \text{sign}(d_{i,x}^0) \mathbf{l}_e + \text{clip}(d_{i,y}^0, -0.35, 0.35) \mathbf{e}_y \right\|.$$

Finally, the sampler draws a radius and transforms the reference-frame offset into world coordinates:

$$r_i \sim \mathcal{U}(0.25, 1.1), \quad \mathbf{c}_i^w = \mathbf{T}_r^w \left(r_i \mathbf{d}_i^{k(i)} \right).$$

`forward_local` keeps the reference rig orientation. The two target-looking families orient the camera to the selected actor-visible target center $\mathbf{p}_e$:

$$\mathbf{z}_i^w = \|\mathbf{p}_e - \mathbf{c}_i^w\|, \quad \mathbf{y}_i^w = \|\mathbf{e}_y - (\mathbf{e}_y^T \mathbf{z}_i^w) \mathbf{z}_i^w\|, \quad \mathbf{x}_i^w = \mathbf{y}_i^w \times \mathbf{z}_i^w.$$

These equations are the mathematical description of the implemented sampler, not a claim that the mixture is optimal. The three-family design has a direct downstream impact: `forward_local` preserves egocentric motion continuity; `target_bearing_local` tests whether moving along the target bearing produces supervised target gain; and `lateral_target_bypass` creates side-step views that may improve occluded target surfaces without leaving the local walking envelope. If the target-aware families do not survive pruning, the resulting rollout dataset degenerates into a forward-only dataset and cannot support a target-conditioned planning claim.

Pruning converts the full shell into a compact valid-action table. A row remains valid only if it lies in the snippet occupancy support, stays clear of the GT mesh, avoids straight-line path collision, and satisfies local egocentric motion limits:

$$\|\mathbf{o}_i\|_2 \leq 1.0 \text{ m}, \quad |\Delta h_i| \leq 0.25 \text{ m}, \quad \max(0, -o_{i,z}) \leq 0.25 \text{ m}, \quad \Delta\psi_i \leq 70 \text{ deg}.$$

The full shell is still retained with `position_id`, `strategy_id`, `mixture_id`, `sampler_probability`, rule masks, debug diagnostics, and invalid-reason bitsets. Invalid candidates are hard-masked constraints with explicit reasons. They are never low-RRR examples, and they must not enter Q_H argmax, softmax, or loss targets. The canonical real config requires at least 15 valid root actions for a 60-row shell, matching the first production gate $\max(12, \text{ceil}(0.25N_q))$. This threshold is a data-support guard: it prevents low-support roots from masquerading as planning evidence, while preflight still reports the blocked roots and per-family failure modes.

3.2.4 Rollout Branch Sampling and Dataset Impact

Rollout generation samples finite branches over the valid candidate table. The canonical thesis profile materializes four recipe families:

Recipe	H	B	Beam	Selection rule
random_valid	1	1	1	Uniform over valid rows.
oracle_greedy	1	1	1	Argmax of oracle target-root gain.
oracle_lookahead	2	2	2	Bounded oracle greedy branches.
temperature_softmax	2	2	2	Softmax over robust oracle scores with $\tau = 1$.

Table 7: Canonical rollout recipes for the real thesis profile. The recipes create replay diversity and oracle-lookahead references; they do not train Q_H by themselves.

For stochastic branches, let s_i be the finite oracle score of valid row i . The robust logit used for temperature-softmax is

$$\ell_i = \frac{s_i - \text{median}(s)}{\text{IQR}(s)\tau}, \quad P(i \mid m_i = 1) = \frac{\exp(\ell_i)}{\sum_{j:m_j=1} \exp(\ell_j)}.$$

The downstream effect of these choices is scientific rather than cosmetic. The target source controls what counts as a supervised task; the candidate mixture controls the support on which Q_H can learn finite-action values; the validity rules decide which actions are admissible; and the branch sampler determines whether the replay store contains only myopic winners or also valid lower-ranked alternatives. A trustworthy thesis dataset must therefore report, per scene and target, selected target counts, valid candidates, valid candidates by `position_id`, invalid reasons by family, selected-family histograms, marginal and cumulative target-root gain, diagnostic state-relative target RRI, and storage/retention settings. Only after these diagnostics show non-degenerate target-aware support should failures or successes be attributed to planning rather than to the data-generation profile.

3.2.5 Target-Specific RRI

Let $C_e(\mathcal{P}_t)$ denote the oracle-only crop of accumulated points to the matched target region. The target error is the target-cropped version of the VIN-NBV RRI

objective [Fra+25]: point-to-mesh accuracy plus mesh-to-point completeness on the crop.

$$\Delta_t^e = d(C_e(\mathcal{P}_t), \mathcal{M}_e^{\text{GT}}) = D_{P \rightarrow M, t}^e + D_{M \rightarrow P, t}^e$$

Area weighting or uniformly sampled target-surface points prevent target-specific RRI from reflecting mesh tessellation density. For reproducibility, the current implementation computes the point-mesh distances through `OracleRRI.score` and `chamfer_point_mesh_batched`, while `aria_nbv.pose_generation.target_counterfactuals` owns the target crop. Empty or unsupported target crops are invalid label cases, not low-RRI samples.

The immediate training reward adapts VIN-NBV’s reconstruction-improvement idea to a target crop and normalizes by the root target error rather than the current error [Fra+25]. This makes equal-horizon rollouts additive against a common root baseline:

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\Delta_0^e + \varepsilon}$$

The finite-horizon return is the discounted sum of those target rewards along a selected counterfactual branch. It is a training target for Q_H , not a claim that the deployed system has an online continuous-control policy:

$$G_t^{(H)} = \sum_{k=0}^{H-1} \gamma^k r_{t+k}^e$$

Endpoint gain is the primary fixed-budget comparison metric because it measures the target quality after the same number of acquisitions for each policy:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

The log-gain variant is retained as a scale-sensitivity ablation, mirroring the broader NBV literature’s use of logarithmic error reduction while keeping the root-normalized endpoint gain as the default thesis metric:

$$J_{e, \log}^{(H)} = \log(\Delta_0^e + \varepsilon) - \log(\Delta_H^e + \varepsilon)$$

$J_e^{(H)}$ is the primary fixed-horizon endpoint metric, $G_t^{(H)}$ is the rollout training return, and $J_{e,\log}^{(H)}$ is only an algebraic scale-sensitivity ablation. The default immediate rollout / Q_H reward is root-normalized by Δ_0^e , so with $\gamma = 1$ its additive return telescopes to endpoint gain up to epsilon under equal-horizon, equal-budget comparisons. The current-error denominator defines state-relative one-step RRI for diagnostics and VIN compatibility only; the final discount and clipping policy remain open protocol parameters.

Open decision: Lock identity-IoU thresholds, ambiguity gap, per-snippet target cap, near-solved-target filtering policy for loaders, clipping, and final gamma policy.

Source: target-selection interview; target-selection autoresearch report

Gate: RQ1/RQ2 protocol freeze

Validation TODO: Regenerate current manifest-backed counts for one-step labels, target-task samples, rollout rows, storage footprint, render cost, invalid reasons, and target-headroom strata before presenting scale or runtime claims.

Source: seminar offline-cache appendix; current rollout-store contract

Gate: M1/M2 data-generation evidence

3.3 Backbone and Scene-Encoder Requirements

The planned value model needs more than a generic 3D feature extractor. A thesis-grade backbone or scene encoder must provide actor-visible target hypotheses, especially OBB detections, and a reusable scene representation for target-, history-, and candidate-conditioned Q_H queries. The current default therefore remains EFM3D / EVL: it is trained for the Project Aria / ASE regime, lifts multi-frame image features into a gravity-aligned 3D volume, uses semi-dense support, and exposes OBB/object evidence [Met25b, Str+24a]. A replacement backbone is only thesis-relevant if it preserves this target-detection contract while improving the conditioning representation used by Q_H .

The central asymmetry is that the historic state is rich and multimodal, while counterfactual states are not. Logged snippets contain calibrated RGB/SLAM streams, poses, semidense points, EVL evidence, and observed or predicted OBBs. Counterfactual successors can reliably add selected geometry, support counts,

visibility history, oracle-rendered labels, and actor-visible summaries derived from those sources; they cannot assume fresh RGB, DINO, semantic, or detector outputs at unvisited candidate poses unless a separate renderable or learned modality generator is introduced and validated. This prevents a model from winning by consuming modalities that exist only for the logged trajectory or only inside the oracle.

The important failure mode is support mismatch. The EVL voxel field is rooted in the snippet evidence and has finite local support; a target OBB, target-local surface cell, or candidate frustum can be physically relevant while partially or fully outside that field. For Q_H , an EVL read therefore contributes both a feature and a coverage diagnostic, not a proof that the target is irrelevant:

$$\omega_{t,i}^{\text{EVL}} = \frac{1}{K} \sum_{k=1}^K \mathbb{1}[x_{t,i,k} \in \mathcal{V}_0^{\text{EVL}}], \quad \mathbf{s}_{t,i}^{\text{EVL}} = \text{Pool}\left(\left\{F_0^{\text{EVL}}(x_{t,i,k}) : x_{t,i,k} \in \mathcal{V}_0^{\text{EVL}}\right\}\right).$$

Here $x_{t,i,k}$ are target-crop, frustum, or target-frustum-intersection query points for candidate i , $\mathcal{V}_0^{\text{EVL}}$ is the root EVL support volume, $\omega_{t,i}^{\text{EVL}}$ is the actor-visible local-support fraction, and $\mathbf{s}_{t,i}^{\text{EVL}}$ is a pooled local evidence token. Low $\omega_{t,i}^{\text{EVL}}$ should usually be a support feature, uncertainty feature, or preflight warning; it becomes hard invalidity only when the row cannot be evaluated or cannot produce a meaningful actor-state transition.

Requirement	Reason	Thesis consequence
OBB-capable target evidence	Target-specific RRI and target-conditioned Q_H need a selectable entity hypothesis, not only a scene embedding.	Backbones without observed/predicted OBB support are scene-feature ablations, not replacements for EVL target evidence.
Aria / ASE domain fit	Egocentric cameras, gravity alignment, semidense support, synthetic-to-real gaps, and ASE OBB taxonomy are part of the data contract.	Prefer EFM3D / EVL unless another model is trained or adapted on ARIA/ASE-style data and reports OBB quality.
Reusable conditioning state	Q_H must condition on target, selected history, support, visibility, and candidate geometry across many candidate rows.	Expose typed query pools rather than one final-pose voxel tensor: target crop, candidate frustum, target-frustum intersection, and directional-history support.
Actor-visible provenance	Historic and counterfactual states have different modality availability.	Every token records source role, frame, feature model, support count, uncertainty, and whether it came from logged observation, selected successor depth, or oracle-only evaluation.
Controlled scale	The thesis should not become a large foundation-model replacement project.	A smaller or cheaper encoder is desirable, but only after OBB quality, queryability, storage, runtime, and invariance tests pass.

Table 8: Hard requirements for a backbone or scene encoder to enter the thesis-core Q_H path. This makes EVL an anchor, not a ceiling. The near-term representation question is whether the current final-pose voxel field hides useful evidence. The first ablations should keep the EFM3D/EVL target detector and add better actor-visible conditioning: semidense/fused point support, compressed DINO-on-point features lifted from logged views, EVL internal or crop reads around target hypotheses, selected-depth successor summaries, and explicit directional visibility memory. These products can give Q_H a broader queryable state without claiming that a new model has solved Aria-native OBB detection.

The following table is a design-contract surface: it classifies representation candidates and their entry gates, rather than reporting final experimental rankings.

Candidate	repre-	Useful role	Boundary before thesis-core
sensation			use
EFM3D / EVL	Default	Aria-native target/ support anchor: OBB evi- dence, lifted visual features, occupancy/free-space support, and local voxel reasoning.	Do not claim full long-horizon mem- ory; report local-volume extent, fi- nal-pose anchoring, and target/can- didate out-of-extent support sepa- rately from oracle invalidity.
Cube R-CNN-style de- tector [Bra+23]	Useful detector baseline or his- torical comparison for single- frame 3D OBB prediction.		Not sufficient as scene encoder; must show ARIA/ASE adaptation and OBB quality before replacing EVL target evidence.
Semidense/fused point bank plus compressed DINO-on-point	First serious scene-condition- ing ablation: point support, uncertainty, observation lin- eage, and local visual evidence are queryable by target and candidate geometry.		Requires feature-cache provenance, compression policy, point-order tests, and joins by stable sample ids; no raw unbounded feature dumps.
EVL internal/crop reads	Tests whether pre-head voxel, neck, or crop evidence around predicted OBBs carries target support lost in final heads.		Must stay actor-visible and store EVL extent, crop frame, grid pose, feature id, and out-of-extent status.
Point/sparse encoders [Qia+22, Tho+19, Wu+24]	Late support-encoder abla- tions when simple target/frus- tum/intersection pools fail.		Need neighborhood, density, se- rialization, frame-canonicalization, storage, and runtime audits; they do not by themselves provide OBB detections.
Explicit/radiance-field state such as 3DGS [Ker+23]	Possible bridge for renderable memory, uncertainty, or miss- ing counterfactual modalities.		Not thesis-core unless trained, stored, and evaluated as actor-visi- ble state rather than oracle target leakage.
Deja View-style looped refinement	Useful recurrence, compute- depth, and failure-diagnostic template for bounded iterative updates.		Not a backbone replacement here; adopt the weight-tied refinement idea only after the fixed-depth can- didate-set path is stable.

Table 9: Backbone and scene-representation ladder. Candidate representations may improve conditioning without replacing the OBB-capable EFM3D/EVL anchor.

3.3.1 Descriptor and Encoding Plan

The descriptor plan follows from this backbone constraint. Q_H should first receive typed, actor-visible query pools rather than a monolithic feature tensor: a target token, selected-history summaries, per-candidate geometry, candidate-target relations, directional visibility memory, and optional feature-bank joins. The storage contract remains canonical: persist IDs, poses, masks, support counts, feature provenance, and oracle labels in replay rows; derive model descriptors in the reader so that new encoders do not rewrite the supervision artifact. This also keeps the historic/counterfactual asymmetry explicit: logged history may contribute lifted visual and semidense evidence, but counterfactual future rows may only use selected successor geometry and validated actor-visible summaries.

This core descriptor plan is intentionally marked as a design contract: every adopted block must stay actor-visible and every escalation needs validation evidence before it becomes a thesis-core claim.

The target descriptor must separate identity, support, and evaluation. Identity says which object the rollout is about; support says what actor-visible evidence currently covers it; evaluation associates that target with hidden GT assets for labels. Conflating these channels creates two opposite errors: low-support targets are silently removed before they can expose non-myopic headroom, or oracle GT crops leak into the actor input. The first descriptor version should therefore expose OBB geometry, class/confidence, relative pose, projected area, semidense support, EVL support, EVL coverage/out-of-extent, and selector provenance as actor features, while storing GT match, GT crop, and target-error fields as label/evaluation metadata only.

Status	Representation	Use and gate
Adopt now	EFM3D/EVL target and support evidence	Aria/ASE-native OBB and surface-regression substrate with posed egocentric streams, semi-dense points, and target hypotheses [Met25b, Str+24a]. Report extent/confidence/support/failures; never promote V0 GT targets or oracle labels to actor-visible state.
Adopt now	Explicit scalar descriptors	Candidate pose, target-relative translation, bearing, range, elevation, R6D orientation, validity reason, strategy id, support counts, projected visibility, and remaining budget [Zho+19]. Row-shuffle, mask-isolation, duplicate-row, and valid-count tests are required.
Adopt now	Canonical replay rows with derived descriptors	Store stable ids, poses, masks, selected transition, reward, successor ids, and provenance; derive descriptor versions in the reader and record source role, frame, feature id, compression, support, and uncertainty.
Ablate next	Query-centric relations and directional memory	Use QCNet-style relative positional encodings for candidate-candidate/history relations and keep target visibility as $\mathbb{S}^2$ memory [e3n25, GML22, JLD24, Zho+23]. Credit only after simpler controls are calibrated.

Table 10: Core descriptor plan for Q_H . This is a design contract, not an empirical ranking.

The escalation table is likewise a controlled backlog surface inside the method chapter. It names plausible representation families while keeping their gates explicit.

Status	Representation	Use and gate
Ablate next	Semidense/fused point bank plus compressed DINO-on-point	Tests broader scene memory than the final EVL volume; EFM3D suggests semidense points and frozen DINO cues matter [Str+24a]. Require stable point/voxel ids, compression policy, feature hash, density tests, and storage/runtime accounting.
Ablate next	EVL internals and target/candidate crop reads	Reads pre-head voxel, neck, or OBB-crop evidence when final heads hide support. Store crop frame, grid pose, EVL extent, feature id, and out-of-extent flag; never extract features from oracle target meshes.
Ablate later	Point and sparse support encoders	PointNeXt, Point Transformer/PTv3, KP-Conv, and sparse convolutions are plausible support encoders, not OBB target detectors [CGS19, Qia+22, Tho+19, Wu+24, Zha+21]. Escalate only after compact query pools bottleneck.
Defer / bridge	Renderable explicit state and recurrence	3DGS-like state may later supply renderable memory, and Deja View-style looped refinement is a compute-depth diagnostic template [Bur+26, Ker+23]. Deja View is not the Q_H planner.
Diagnostic only	Cube R-CNN-style OBB detector	Useful as a single-frame detector reference, but EFM3D reports off-the-shelf Cube R-CNN generalizes poorly to egocentric Aria data [Bra+23, Str+24a]. It is not reusable scene-conditioning state.

Table 11: Escalation candidates for scene encodings and backbone alternatives. Each candidate needs actor-visible training, storage, runtime, and ablation evidence before thesis-core use.

The minimal descriptor schema can be summarized as a typed row vector before the table expands the field roles:

$$\mathbf{x}_{t,i} = \text{concat}(\mathbf{z}_e, \mathbf{p}_{t,i}, \mathbf{g}_{t,i}, \varphi_{\text{valid}}(m_{t,i}, \rho_{t,i}), \mathbf{H}_t),$$

with target descriptor $\mathbf{z}_e$, candidate-pose token $\mathbf{p}_{t,i}$, candidate-geometry token $\mathbf{g}_{t,i}$, hard-mask/reason encoding φ_{valid} , and selected-history summary $\mathbf{H}_t$. This symbolic view keeps the table below as a provenance checklist rather than the model definition.

Descriptor block	Fields	Failure prevented
Target token	Observed or predicted OBB center, extents, orientation, class probabilities, detector confidence, selector rank, projected visibility, semidense/EVL support, EVL coverage/out-of-extent, and source mode.	Separates V1 actor-visible target input from GT-EVAL matching and prevents GT OBB leakage into deployable policy inputs.
Candidate self token	Candidate pose in root, target, and current-camera frames; distance, bearing, elevation, optical-axis alignment, remaining horizon, sampler strategy, validity mask, and reason code.	Makes the finite action row interpretable and keeps invalidity a hard mask rather than a low-reward training example.
Candidate-target relation	Frustum-target overlap, projected target area, expected support change, target-local approach direction, visibility novelty, and optional EVL/crop support read.	Forces the model to condition on the chosen target instead of learning scene-level coverage shortcuts.
Candidate-candidate and history relations	Relative poses, angular separation, duplicate/near-duplicate indicators, selected-view directional memory, and valid-count features.	Supports permutation-equivariant interaction while exposing attention-normalization shortcuts and duplicate-row artifacts.
Feature-bank joins	Stable point or voxel id, feature-model id, compression version, observation count, frame lineage, uncertainty, and out-of-extent status.	Prevents silent mixing of logged RGB/DINO features with counterfactual rows that only have selected successor geometry.

Table 12: Minimum descriptor schema before heavier scene encoders are evaluated. Every field is either actor-visible or explicitly stored as oracle-only label/evaluation metadata outside the policy input.

Validation TODO: Before promoting any feature-bank, point/sparse, radiance-field, or recurrent scene encoder to thesis-core status, run EVL-extent, source-dropout, density, row-shuffle, mask-isolation, storage, and runtime ablations against the simpler descriptor controls.

Source: autoresearch thesis-lit-review Iterations 16, 24, 25, and 26; local EFM3D and Deja View literature refresh

Gate: representation ablation evidence

Validation TODO: Resolve the current wording/code tension around outside-EVL extent: some rollout contracts list outside-EVL support as a hard reason, while current VIN evidence treats low EVL coverage as a diagnostic/support feature. The thesis-core rule should be: infeasible pose, missing oracle/evaluation sample, or empty target crop is hard invalidity; low local EVL support alone is a model feature unless it blocks evaluation.

Source: docs/contents/thesis/questions.qmd:21; aria_nbv/aria_nbv/vin/experimental/model.py:848; aria_nbv/aria_nbv/vin/model_v3.py:1465

Gate: rollout invalidity audit before first Q_H training run

3.4 Candidate and Replay Contract

Each decision state carries a finite candidate table $\mathcal{Q}_t$, hard mask $\mathbf{m}_t$, invalid-reason vector $\boldsymbol{\rho}_t$, and the target descriptor $\mathbf{z}_e$. It also stores selected-view history and remaining budget. The admissible action is a valid candidate row index:

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad \mathcal{A}_t = \{i \in \{1, \dots, N_q\} : m_{t,i} = 1\}$$

Selecting a candidate means choosing a valid index $a_t = i \in \mathcal{A}_t$ for the transition. Oracle rendering follows the repository’s PyTorch3D depth-rendering path, so camera-frame and rasterizer conventions are part of the label contract rather than model input [Met25c]. All valid candidates may be rendered at the oracle layer to score one-step labels, while the rollout writer separately persists selected/parent depth at a canonical configured resolution as actor-history state for successor Q_H encoders.

After selection, acquired geometry is added to the current geometry:

$$\mathcal{P}_{t+1} = \mathcal{P}_t \cup \mathcal{P}_{t,i}^{\text{cand}}.$$

The next candidate table $\mathcal{Q}_{t+1}$ is regenerated from updated geometry, selected-view history, and remaining horizon metadata with the same logged mixture families, while root local EVL evidence remains fixed unless a later ablation explicitly recomputes it. The current target-conditioned mixture vocabulary contains forward/local candidates, target-bearing candidates, lateral target-bypass candidates, bounded orientation jitter, and per-row strategy provenance. The older radial free-shell sampler from the seminar paper is retained as a historical

upper-bound or stress ablation, not as the default target-conditioned candidate distribution.

Candidate provenance is a model input only through typed scalar or embedding channels. The row stores `strategy_id`, `position_id`, `mixture_id`, `sampler_probability`, target-distance/bearing diagnostics, motion-realism diagnostics, and invalid-reason bits. The training reader may embed these as candidate-family tokens, but the model must still pass row-shuffle and duplicate-row tests: the family label explains how the row was sampled, not an ordering prior.

Candidate order has no semantics, so shuffled-candidate evaluation is required. Candidate orientation uses a continuous 6D rotation code [Zho+19], but accumulated visibility is a directional memory over $\mathbb{S}^2$, not a sum of R6D vectors. For a target point or voxel center $\mathbf{v}$ and a previously selected camera center $\mathbf{c}_k$, the observed direction is

$$\mathbf{d}_k(\mathbf{v}) = \frac{\mathbf{c}_k - \mathbf{v}}{\|\mathbf{c}_k - \mathbf{v}\|_2}$$

Here $\mathbf{v} \in \mathbb{R}^3$ is a target-local point or voxel center, $\mathbf{c}_k \in \mathbb{R}^3$ is the center of a selected camera, and $\mathbf{d}_{k(\mathbf{v})} \in \mathbb{S}^2$ is the unit ray from the target-local support location toward that selected view. The scalar weight $w_{k(\mathbf{v})}$ can encode whether view k observed, supported, or improved that target-local region. The planned actor-visible feature branch stores history either as low-order spherical-harmonic coefficients [e3n25]

$$\mathbf{h}_{\text{dir}}(\mathbf{v}) = \sum_{k < t} w_k(\mathbf{v}) \mathbf{Y}_L(\mathbf{d}_k(\mathbf{v}))$$

or as a second-moment summary over unit directions,

$$\mathbf{M}_{\text{dir}}(\mathbf{v}) = \sum_{k < t} w_k(\mathbf{v}) \mathbf{d}_k(\mathbf{v}) (\mathbf{d}_k(\mathbf{v}))^\top$$

from which the candidate can read a directional novelty score. The numerator projects the candidate direction through the accumulated second-moment matrix; the trace-normalized complement is high when the candidate approaches the target from an underrepresented direction:

$$\nu_{t,i}^{\text{dir}}(\mathbf{v}) = 1 - \frac{(\mathbf{d}_{t,i}(\mathbf{v}))^\top \mathbf{M}_{\text{dir}}(\mathbf{v}) \mathbf{d}_{t,i}(\mathbf{v})}{\text{tr}(\mathbf{M}_{\text{dir}}(\mathbf{v})) + \varepsilon}$$

The directional-memory diagram in Figure 4 keeps this branch separate from generic pose tokens: selected views first accumulate target-local directional evidence, and each candidate then queries whether it sees the target from a genuinely new direction.

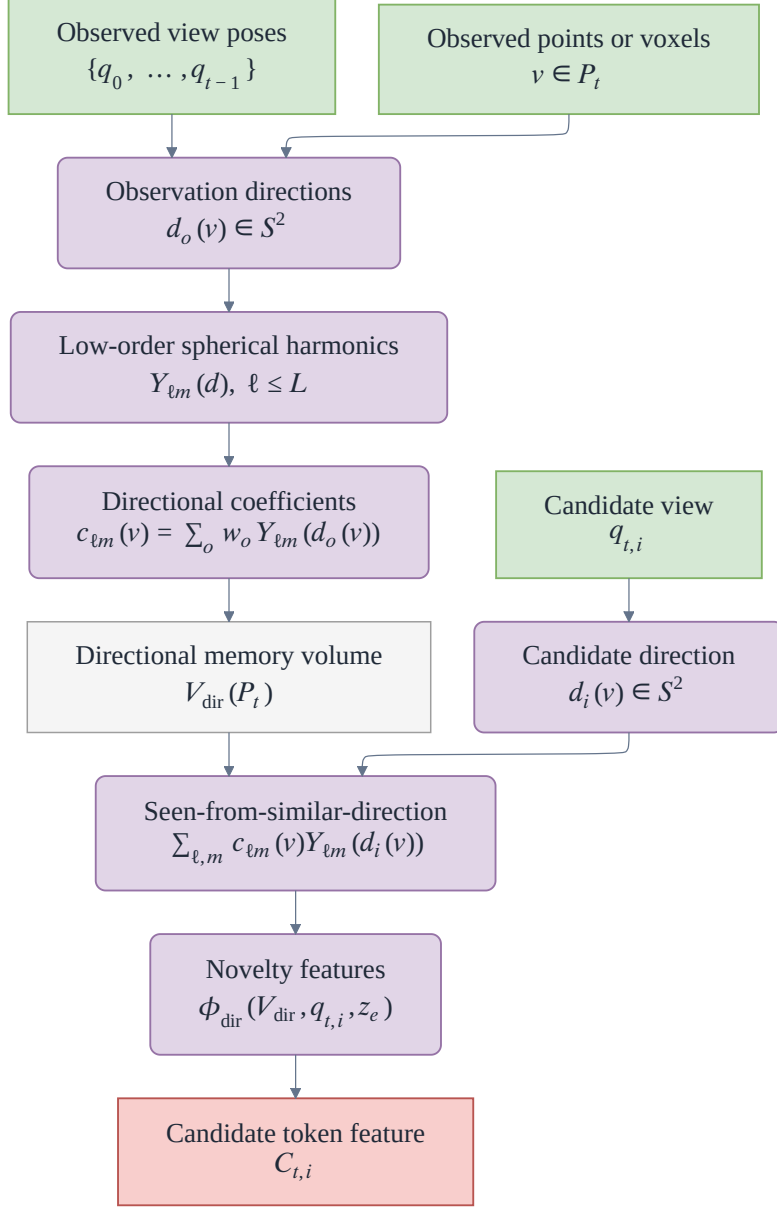


Figure 4: Actor-visible directional memory for target-local view novelty. The figure shows a planned descriptor branch, not an implemented performance result: selected view directions over observed points or voxels are summarized as low-order directional coefficients, and each valid candidate reads the memory to produce a candidate token feature for Q_H .

The minimum replay row contains scene/snippet/target/step identifiers, counterfactual state, target descriptor, candidate table, masks, invalid reasons, selected action, target reward, successor state, successor candidates, successor masks, and

policy/seed/sampler metadata. This row reproduces the mask, selected transition, value target, and oracle re-evaluation.

3.5 Architecture Contract and Geometric Acceptance Tests

The geometric-learning rationale in Section 2.2 becomes acceptance tests over the replay fields here. The planned architecture is not justified by adding geometric modules until the model looks sophisticated. Each module must answer a specific symmetry or provenance requirement of the finite-candidate problem. Candidate rows form an unordered set, so the learned value map must be permutation equivariant at the row level: reordering $\mathcal{Q}_t$ may reorder Q_H outputs, but it must not change the value attached to a physical candidate. Invalid and padded rows are constraints, so mask isolation is part of the architecture contract rather than a post-processing detail. Candidate, target, and current-history geometry must be encoded in local frames so that arbitrary world-frame origin or yaw conventions do not become shortcuts. At the same time, the task is gravity aligned and egocentric; the thesis should not claim full $SO(3)$ or $SE(3)$ equivariance for the whole system.

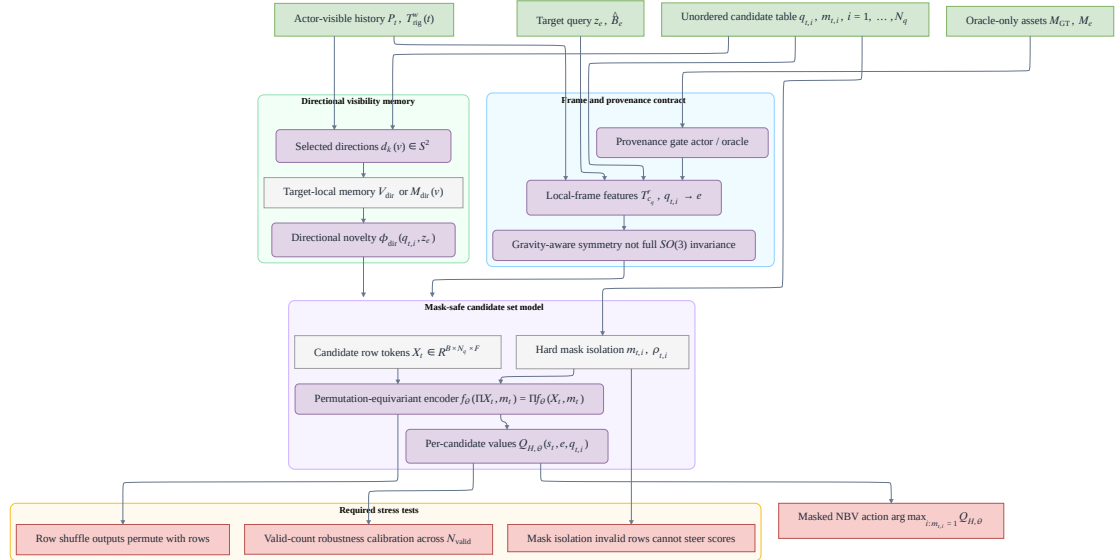


Figure 5: Minimum symmetry and provenance contract for the finite-candidate Q_H model. The contract requires row-level equivariance, mask isolation, local-frame geometry, target-local directional memory, and oracle/actor provenance gates; it does not claim exact global $SE(3)$ equivariance.

The first model family should therefore use scalar invariant and local-frame relative features before heavier equivariant tensor machinery. A candidate-local relation such as $\mathbf{T}_{c_q}^r$ or a target-relative bearing is a deliberate gauge choice: it removes irrelevant global-coordinate dependence while preserving the yaw, elevation, distance, and approach-direction signals needed for visibility. Directional history is a separate object. A selected view direction belongs on $\mathbb{S}^2$ and should be stored as a target-local histogram, second-moment matrix, or low-order spherical-harmonic memory rather than being merged into generic pose features. This separation protects the interpretation of an ablation: a QCNet-style relative positional bias tests candidate-candidate geometry, while directional memory tests whether the target has already been observed from similar directions [e3n25, Zho+23].

The architecture acceptance tests are as important as validation loss. Row-shuffle tests must satisfy $f_\theta(\Pi X_t, m_t) = \Pi f_\theta(X_t, m_t)$ up to numerical tolerance for every per-candidate output used by selection. Mask tests must show that invalid rows cannot alter valid scores except through explicit valid-count or support features. Valid-count and duplicate-row stress tests check whether attention normalization has corrupted absolute target-specific RRI calibration. Only after the independent scorer and DeepSets controls pass these tests should masked Set Transformer interaction, Fisher/SCONE overlap bias, QCNet-style local relative positional encoding, or EGNN-style candidate graphs be credited as architectural gains [GML22, JLD24, Lee+19, SHW21, Zah+17].

The token design follows the same separation of concerns. State tokens summarize what is known before choosing the next view; candidate tokens describe one admissible or invalid finite action; relation tokens describe target/candidate/history geometry; label tokens do not enter the actor graph. This keeps the first model educationally simple: start with a per-candidate scorer, add pooled set context, then add masked attention only when interaction is needed.

Token role	Content	Invariant / gate
Target token	Actor-visible OBB geometry, class/confidence, projected area, semidense/EVL support, relative target pose, and support coverage.	No GT crop, GT identity, or oracle target error in actor input.
Candidate-row token	Pose in root/current/target frames, R6D orientation, strategy/position ids, motion diagnostics, mask/reason bits, and candidate-local support reads.	Row permutation equivariance and hard-mask isolation.
Scene/support token	Root EVL local evidence, semidense/fused point support, selected-depth successor summaries, optional compressed DINO-on-point features.	Actor-visible provenance, feature hash/compression version, support count, and EVL extent status.
History token	Selected-view poses, remaining budget, accumulated target-local directional memory, and selected successor depth metadata.	No all-candidate oracle renders; only selected successor geometry updates history.
Relation token	Candidate-target frustum overlap, target-frustum intersection support, candidate-candidate relative pose, and QCNet-style local RPE.	Ablation-gated after independent and DeepSets controls.
Label/evaluation token	GT target crop, target RRI, root-normalized target gain, oracle endpoint metric, and TD target.	Stored in replay/q_h views only; never fed into the actor graph.

Table 13: Token ownership for the first finite-candidate Q_H architecture. The table defines what may enter the actor model, what remains an ablation, and what is label/evaluation only.

Research TODO: Treat Fisher/SCONE overlap attention and QCNet-style relative encodings as ablation hypotheses until row-shuffle, mask-isolation, and paired oracle policy evidence show they improve target-specific endpoint gain over simpler controls.

Source: autoresearch thesis-lit-review report; docs/contents/theory/candidate_view_dependence.qmd

Gate: A2/A3/A4 architecture ablation tables

3.6 Finite-Candidate Value Model

The value-model hypothesis is that a masked finite-candidate model can recover positive oracle-lookahead headroom from actor-visible state. Q_H maps each valid candidate row to a finite-horizon value using actor-visible scene, target, selected-history, budget, candidate, mask, and reason-code features. Its outputs select

actions, but thesis evidence comes from oracle re-scoring of the selected trajectories.

The model class follows the structure of the decision problem. The action space is a masked finite set of candidate views, each defined relative to a target, selected history, and partially observed geometry. Geometric deep learning supplies vocabulary for these regularities without committing the thesis to a full equivariant tensor network [Bro+21]. Candidate-row permutation requires equivariant per-candidate outputs; local camera and target frames reduce dependence on global coordinates; $\mathbb{S}^2$ visibility memory records where the target has already been observed; and the target record acts as the query that determines which reconstruction errors matter.

The architecture critique is therefore simple: the first serious model should not be a monolithic transformer over every available tensor. The clean object is a calibrated one-step target utility field plus a masked residual correction over the valid candidate set. The one-step field owns absolute target-gain calibration; the set module owns only context-dependent advantage, redundancy, and finite-horizon effects. This prevents unrelated sampled rows from changing the physical one-step label attached to a candidate, while still allowing non-myopic effects from candidate regeneration, selected-history geometry, occlusion, and support overlap.

$$b_{\psi,i} = f_{\psi}^{\text{1-step}(s_t^{\text{eff}}, z_e, q_{t,i})}, \quad A_{\theta,i} = g_{\theta}(\mathbf{x}_{t,i}, \{\mathbf{x}_{t,j}\}_{j \in \mathcal{A}_t}, h_t),$$

$$Q_{H,\theta,i} = b_{\psi,i} + A_{\theta,i} - \frac{1}{|\mathcal{A}_t|} \sum_{j \in \mathcal{A}_t} A_{\theta,j}.$$

The zero-mean residual form is not required forever, but it is the most defensible first architecture: if set interaction helps, it improves ranking through relative candidate context; if it does not, the model collapses to the calibrated myopic scorer rather than corrupting the target-RRR scale. Recent object-centric view-planning work reinforces this separation: target-centric visibility and feasibility should be explicit scoring factors, while difficulty, reachability, budget, and object

saturation must be reported separately because they can change planner rankings and failure modes [Hu+26, Pan+26].

Design principle	Adopt in Q_H	Reject or defer
Calibrated absolute field	Keep a one-step target-gain head that can be evaluated candidate-by-candidate under fixed masks.	Letting attention over unrelated rows refine the absolute immediate-RRI label.
Residual set context	Use DeepSets or masked Set Transformer context as a zero-mean advantage correction over valid rows.	A pooled scene embedding with no per-row path, or unmasked invalid-row attention.
Typed relative geometry	Encode target-current-candidate-history relations in local frames; use QCNet-style RPE as an ablation.	Claiming full global SE(3) equivariance for an egocentric, gravity-aligned, frustum-limited task.
Directional visibility	Represent selected history on S^2 and keep it distinct from generic pose tokens.	Collapsing target-local observability into a scalar distance or sampler family prior.
Support and difficulty reporting	Report EVL extent, semidense support, reachability, validity, budget, and target saturation bins.	Using support, Fisher/coverage proxies, or target area as the thesis reward instead of oracle target-RRI.
Late geometric transformers	Treat EGNN, SE(3)-Transformer, and GATr-style equivariant modules as support-encoder or candidate-graph ablations.	Making exact equivariance the first thesis claim before the replay/mask/target-RRI contract is stable [Bre+23, Fuc+20, SHW21].

Table 14: Architecture critique distilled into a conservative design order. The clean first model preserves a calibrated candidate-local field and allows set/geometric context to explain only residual advantage under explicit masks and evaluation gates.

Object	Structure	Bias	Implementation role
candidate table $\mathcal{Q}_t$	finite set	permuting rows per- mutes per-candidate Q_H outputs	independent candidate MLP, pooled DeepSets residual context, and masked Set Transformer residual context [Lee+19, Zah+17]
camera, target, SE(3) / and history poses	gravity-aligned local frames	relative geometry re- duces dependence on arbitrary global frame choice	relative target/history pose features, R6D rota- tions, QCNet-style candi- date-local RPE; EGNN- style graph as abla- tion [SHW21, Zho+19, Zho+23]
directional visibil- ity	$\mathbb{S}^2$	encode which directions have already observed target-local cells	second-moment memory by default; histogram or low-order spherical har- monics as ablations [e3n25]
target identity	selected entity / OBB query	target defines the query frame for value estima- tion	target-task descriptor and predicted-OBB crop; GT crops remain labels/evalu- ation

Table 15: Symmetry contract for the value model. Added structure is evaluated by oracle-scored endpoint target quality under the paired headroom protocol.

The one-step target scorer is adapted to counterfactual rollout rows rather than reusing the seminar VIN checkpoint unchanged. It remains myopic and predicts immediate target-specific RRI evidence for each candidate; the seminar VINv3 scorer is therefore a control architecture and implementation substrate, not already a target-conditioned finite-horizon Q_H result. Q_H is residual around this calibrated base, with the dueling residual head as the canonical value definition. The myopic scorer uses the CORAL ordinal-regression interface of Cao et al. and the `coral-pytorch` layer/loss implementation, adapted to ARIA-NBV’s skewed oracle RRI labels [CMR19, Ras25]:

$$\begin{aligned}
p_{t,i,k}^{\text{CORAL}} &= \sigma(o_{t,i,k}^{\text{CORAL}}), \quad k = 0, \dots, K-2 \\
\pi_{t,i,k}^{\text{CORAL}} &= p_{t,i,k-1}^{\text{CORAL}} - p_{t,i,k}^{\text{CORAL}}, \quad p_{t,i,-1}^{\text{CORAL}} = 1, \quad p_{t,i,K-1}^{\text{CORAL}} = 0 \\
\hat{r}_{\psi}^e(s_t^{\text{cf0}}, \mathbf{z}_e, q_{t,i}) &= \sum_{k=0}^{K-1} \pi_{t,i,k}^{\text{CORAL}} u_k
\end{aligned}$$

ARIA-NBV’s adaptation is in the binning and decoding around that interface. Continuous oracle target gains are fitted to empirical quantile edges $\tau_1 \leq \dots \leq \tau_{K-1}$, and each sample receives the ordinal label $y = \sum_{j=1}^{K-1} \mathbb{1}[r^e > \tau_j]$, matching the repository’s `RriOrdinalBinner`. CORAL levels are threshold indicators $\ell_k = \mathbb{1}[y > k]$ for $k = 0, \dots, K-2$, matching `ordinal_labels_to_levels`. The code then decodes logits both as cumulative probabilities $P(y > k) = \sigma(o_k)$ and as a ranking proxy $E[y] = \sum_k \sigma(o_k)$; when calibrated bin representatives are available, the expectation over u_k maps the ordinal distribution back to target-gain units. This preserves the VIN-NBV ordinal-ranking precedent while making calibration, bin drift, and residual Q_H recovery explicit ARIA-NBV diagnostics.

Training is staged to preserve the residual interpretation: train and calibrate $\hat{r}_{\psi}^e$, then freeze or slow-finetune it while fitting residual Q_H , and finally ablate whether end-to-end fine-tuning improves oracle-evaluated policy performance. This also keeps the educational story clean: VINv3 answers “which single candidate looks good now?”, while Q_H answers “which first action has the best bounded downstream target gain under the same finite candidate and mask contract?”.

The first Q_H implementation should start from implemented VIN/EVL heads plus semi-dense or fused geometry. A richer queryable feature bank is a planned representation ablation, not a current persisted cache schema:

$$\Phi_t^{\text{scene}} = (\mathbf{X}_t^{\text{pt}}, \mathbf{F}_t^{\text{DINO@pt}}, \mathbf{E}_0^{\text{EVL-local}}, \mathbf{O}_t^{\text{pred}}, \mathbf{M}_{\text{dir}})$$

Candidate rows query the target crop, candidate frustum, and their intersection:

$$\begin{aligned}
\mathbf{z}_e &= \text{Pool}_{\mathbf{p}_j \in \hat{\mathbf{B}}_e} \mathbf{x}_j^{\text{pt}} \\
\mathbf{z}_i^{\text{fr}} &= \text{Pool}_{\mathbf{p}_j \in \text{Frustum}(q_{t,i})} \mathbf{x}_j^{\text{pt}} \\
\mathbf{z}_{e,i}^{\text{cap}} &= \text{Pool}_{\mathbf{p}_j \in \hat{\mathbf{B}}_e \cap \text{Frustum}(q_{t,i})} \mathbf{x}_j^{\text{pt}}
\end{aligned}$$

The candidate encoder is permutation-equivariant:

$$\{\mathbf{u}_{t,i}\}_{i=1}^{N_q} = E_{\text{set}}\left(\{\text{concat}(\mathbf{x}_{t,i}, \mathbf{T}_e, \mathbf{H}_t)\}_{i=1}^{N_q}, \mathbf{m}_t\right)$$

The candidate-query architecture in Figure 6 expands the same contract into the architecture used for the thesis hypothesis and ablation ladder.

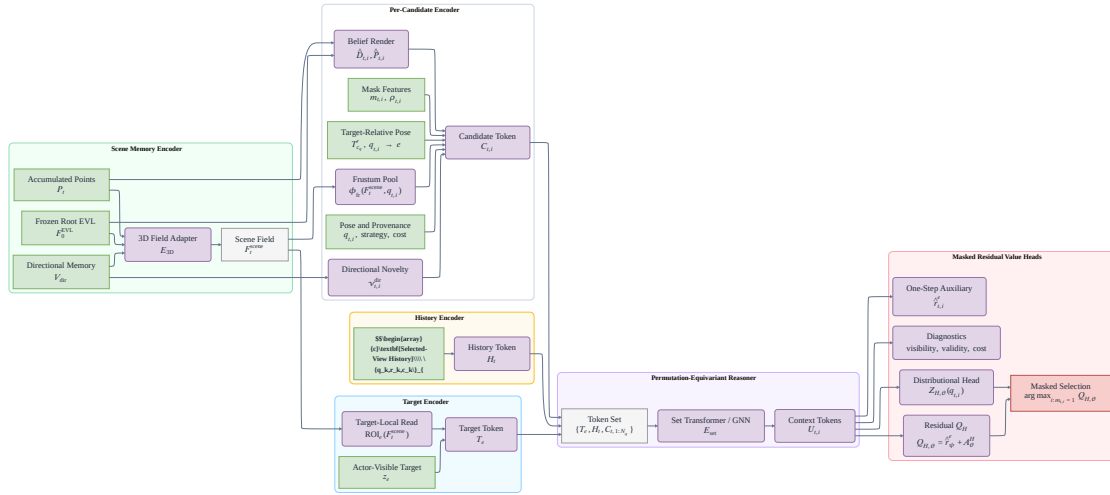


Figure 6: Candidate-query Q_H architecture sketch. Actor-visible EVL evidence, accumulated geometry, target descriptors, selected-history summaries, candidate geometry, masks, and directional memory feed a permutation-equivariant set reasoner; residual finite-horizon values, one-step auxiliary scores, and diagnostics are then decoded only over valid candidate rows. Oracle labels supervise training and evaluation outside this actor-input graph.

No-interaction candidate MLP scoring and pooled DeepSets aggregation are required baselines before attributing gains to masked Set Transformer interaction or QCNet-style RPE. For immediate target-specific RRI, the physical oracle label of candidate q_i does not change when unrelated rows are added to Q_t . Candidate interaction can therefore corrupt absolute calibration if it replaces the independent scorer. The safer myopic ablation uses candidate context only as a zero-mean relative advantage correction.

The value head is a dueling residual decomposition over valid actions [Wan+16]:

$$Q_{H,\theta}(s_t^{\text{cf0}}, z_e, q_{t,i}) = \hat{r}_\psi^e(s_t^{\text{cf0}}, z_e, q_{t,i}) + V_\theta(s_t^{\text{cf0}}, z_e, \mathbf{H}_t) + A_{\theta,i}^H - \frac{1}{|\mathcal{A}_t|} \sum_{j \in \mathcal{A}_t} A_{\theta,j}^H$$

Model role	Content
Hypothesis model	adapted target-conditioned myopic scorer, zero-mean residual set context, residual dueling Q_H , hard masks/reasons, matched-budget oracle re-scoring
Required controls	independent candidate MLP, calibrated myopic scorer, and pooled DeepSets residual context over valid candidate rows
Ablations	QCNet-style candidate-local RPE, Fisher/SCONE support-overlap attention bias, $\mathbb{S}^2$ memory variants, EGNN-style candidate graph, privileged-teacher distillation, distributional Q_H heads
Architecture ladder	A0 independent scorer; A1 pooled DeepSets residual context; A2 masked Set Transformer residual context; A3 query-local relative bias; A4 Fisher/SCONE overlap bias; A5 residual dueling Q_H
Bridges	Hestia-style target-then-pose policies, online discrete interaction, external mesh/oracle-compatible simulators, sparse/point backbones

Table 16: Value-model hypothesis, controls, and ablations. Dense GT candidate renders may supervise later ablations, while learned policy inputs use the configured state and target-task descriptors.

Implementation TODO: Confirm which architecture ladder levels are implemented, planned, or deferred once the final code state is frozen.
Source: advisor handout; proposal method
Gate: method implementation audit

4 Evaluation

The thesis objective is a leakage-safe target-aware NBV stack whose selected views reduce target point-mesh error under a fixed acquisition budget. The endpoint metric $J_e^{(H)}$ is primary, and the additive return $G_t^{(H)}$ is the training target for value learning.

4.1 Objectives and Hypotheses

The first aim defines target-specific oracle RRI while keeping target selection and model input actor-visible. V1 uses observed or predicted target descriptors $\mathbf{z}_e$. GT crops, boxes, meshes, and all-candidate renders are restricted to labels, upper bounds, and evaluation. The required evidence is target eligibility, match score, unmatched/ambiguous counts, endpoint $J_e^{(H)}$, separate scene RRI, and acquisition cost.

The render-path boundary in Figure 7 makes this restriction explicit for render-derived training signals.

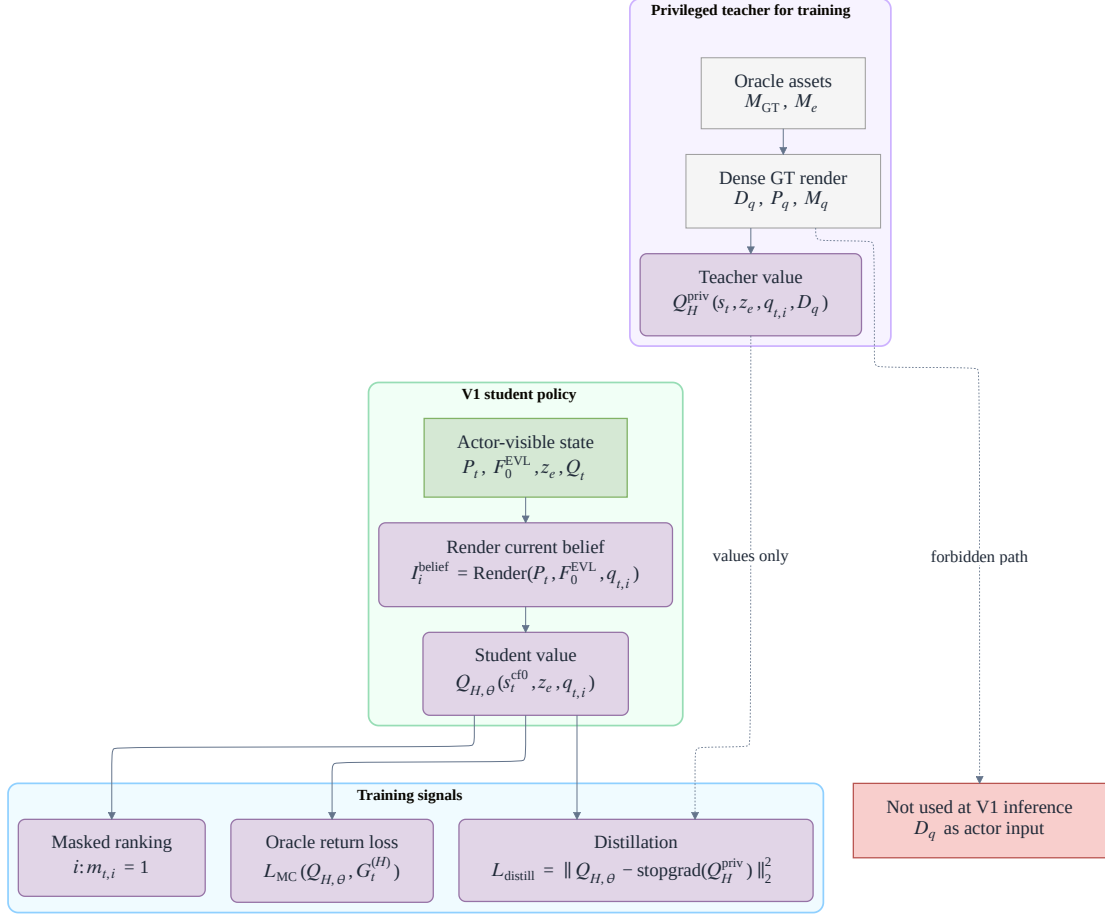


Figure 7: Teacher/student render path for leakage-safe training. The student branch consumes actor-visible state and current-belief render products, while privileged GT meshes, target crops, and dense candidate renders may produce oracle returns, teacher values, or distillation targets only. Dense GT candidate depth is therefore label or teacher evidence, not a V1 actor input.

The second aim trains a VIN-style myopic scorer over the same candidate table that later feeds Q_H . The scorer predicts target RRI from actor-visible scene, target, and candidate features. It is the required learned one-step control, evaluated by rank correlation, top- k oracle hit rate, calibration, selected-candidate oracle RRI, target visibility, invalid fraction, and grouped failures.

The third aim first estimates whether bounded oracle lookahead has headroom over one-step oracle greedy:

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

Only if this headroom is positive is Q_H expected to recover part of it from offline rollout traces:

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

Success is measured by oracle-rescored selected actions, not predicted values. If oracle lookahead itself has little headroom, the thesis reports that the current objective and candidate distribution are effectively myopic. If lookahead has headroom but Q_H fails to recover it, the analysis reports whether the limiting factor is target observability, candidate support, rollout coverage, reward definition, or model capacity.

Claim	Primary evidence	Decision rule
Target utility	$J_e^{(H)}, G_t^{(H)}$, scene RRI, cost	$J_e^{(H)}$ decides endpoint quality; $G_t^{(H)}$ trains and ranks rollouts.
Input safety	actor-visible z_e , match score, support, leakage checks	GT is label/evaluation only for the V1 result.
Myopic control	target-rank metrics, selected-candidate oracle RRI, calibration	One-step target scoring is the comparator for Q_H , not the final policy claim.
Planning headroom	Δ_{look} and recovered fraction η_Q	Q_H is meaningful only relative to measured oracle-lookahead headroom.

Table 17: Objective-to-evidence matrix.

4.2 Learning Objective and Replay Evidence

The learned value model is interpreted only after the myopic scorer passes ranking, calibration, and oracle-selected rollout checks, and after the replay store passes support, mask, seed, and successor-table checks. These gates make the evaluation contract a scientific guardrail rather than a post-hoc reporting checklist. Seminar-paper diagnostics such as scene-level RRI ranking, CORAL bin behavior, and offline-cache storage are reported as historical substrate evidence unless they are regenerated under the target-task sampler and rollout-store protocol.

The replay pipeline in Figure 8 is the operational boundary for this interpretation: all-candidate labels can train and calibrate the one-step scorer, while Q_H evidence comes only from selected-transition rows whose successor state and successor candidate mask are reproducible.

Data product	Purpose	Minimum evidence
one-step all-candidate labels	train and calibrate $\hat{r}_\psi^e$	rank correlation, top- k oracle hit, calibration
paired greedy/lookahead roots	measure Δ_{look}	same root, target, seed, N_q , horizon, and validity masks
stochastic support traces	avoid fitting Q_H only on greedy states	policy entropy, unique selected chains, target visibility
invalid near-misses	stress hard masks and failure modes	invalid-reason distribution and with/without-mask diagnostics
target and candidate strata	test support across target difficulty and candidate families	class/support/area/occlusion bins, per-strategy RRI histograms, successor-table availability

Table 18: Rollout-support coverage. Target, candidate, validity, and successor-table gaps are reported before interpreting value-model differences.

The one-step scorer is trained on all valid candidate rows with oracle immediate target-specific RRI labels. The bootstrapped Q_H backup is trained only on expanded transition rows for which the selected action, successor counterfactual state, successor candidate table, successor mask, and terminal flag are materialized. If all-candidate successor expansion is unavailable, non-expanded candidates contribute myopic supervision but not bootstrapped finite-horizon targets.

Evidence surface	Seminar evidence role	Current thesis gate
Oracle RRI labeler	Implemented scene-level depth-render, fusion, and point-mesh scoring substrate.	Target-cropped labels, empty-crop in-depth-render, fusion, and identity-match audit validity, and identity-match audit pass on current manifests.
Candidate sampling	Legacy free-shell candidate generator and pruning diagnostics.	Target-conditioned mixture reports strategy counts, invalid reasons, headroom strata, and hard-turn diagnostics.
CORAL scorer	One-step ordinal RRI calibration and expected-RRI interface.	Target scorer passes ranking/calibration before residual Q_H is interpreted.
Offline stores	Immutable one-step VIN payload proves feasibility of materialized labels.	Selected-transition rollout store reloads successor tables, masks, seeds, storage, and replay integrity.
Architecture	VINv3/EVL myopic scorer is the implemented control.	A0-A5 ablation ladder is compared by paired oracle-rescored endpoint gain.

Table 19: Substrate-to-thesis evidence matrix used to avoid treating historical seminar results as final target-Q_H claims.

The finite-candidate value model decodes actions only over valid candidate tokens:

$$\mathbf{u}_{t,i} = (\text{Transformer}_\theta(\mathbf{X}_t^{\text{cand}}))_i$$

$$a_t^\theta = \operatorname{argmax}_{i:m_{t,i}=1} Q_{H,\theta}(s_t^{\text{cf0}}, \mathbf{z}_e, q_{t,i})$$

The initial backup is fitted masked Double-Q [HGS15]. Here $d_t = 1$ at horizon termination, budget termination, or when no valid successor action exists:

$$i^\star = \operatorname{argmax}_{i:m_{t+1,i}=1} Q_{H,\theta}(s_{t+1}^{\text{cf0}}, \mathbf{z}_e, q_{t+1,i})$$

$$y_t = r_t^e + \gamma(1 - d_t)Q_{H,\theta^-}(s_{t+1}^{\text{cf0}}, \mathbf{z}_e, q_{t+1,i^\star})$$

The replay dataset $\mathcal{D}$ contains selected-action transition rows with state, action, immediate target reward, successor state, validity masks, and terminal flags:

$$\mathcal{L}_Q(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(s,a,r,s') \in \mathcal{D}} m_{t,a} (Q_{H,\theta}(s_t^{\text{cf0}}, \mathbf{z}_e, q_{t,a}) - y_t)^2$$

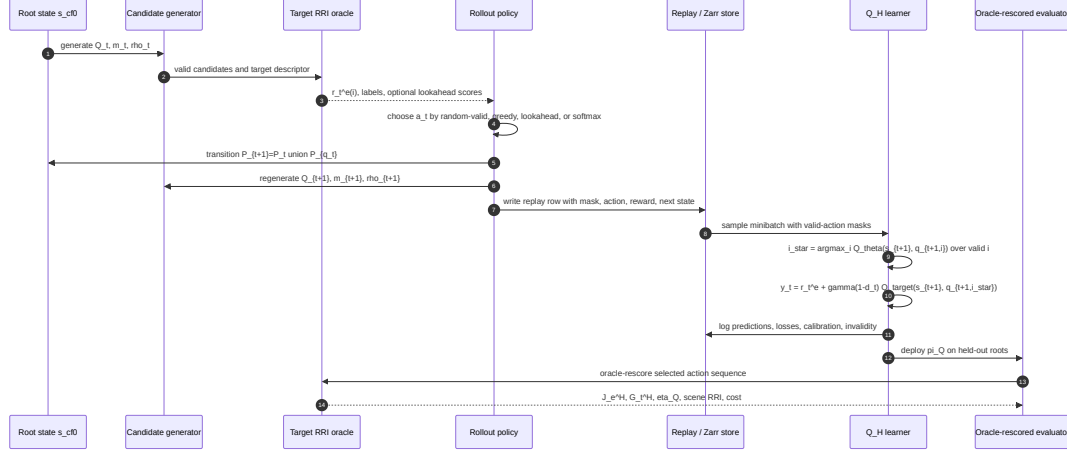


Figure 8: Selected-transition replay contract for Q_H . All valid candidates can receive one-step oracle target labels, but bootstrapped finite-horizon targets require materialized selected actions, successor counterfactual state, regenerated successor candidate tables, masks, terminal flags, and masked Double-Q targets.

4.3 Policy Comparison

All selected actions are oracle-evaluated under the same acquisition and candidate budgets. Equal budget means equal selected-view horizon H , candidate count N_q , candidate-generation distribution, and validity constraints; path length, runtime, and oracle evaluation count are reported separately. Coverage is reported against the planned full scale bar of 100 GT mesh ASE scenes and 4,608 snippet windows from the current thesis contract, or against an explicit scene-level held-out subset if scale is blocked [Met25a]. Final splits are scene-level; sample-level splitting across snippets from the same scene is not valid for final claims.

Validation TODO: Replace planned scale bars with generated manifest statistics for the current one-step store and rollout store; historical seminar subset counts are not current thesis evidence unless regenerated. Source: docs/typst/seminar_paper/sections/04-dataset.typ; docs/typst/seminar_paper/sections/09a-evaluation.typ; current dataset manifests
Gate: final experiment tables

Policy	Actor	in-	GT	deci-	H	Role
	put			sion		
π_{rand}	yes		no		1	lower reference over valid candidates
$\pi_{\text{learned-1}}$	yes		no		1	myopic learned target scorer
$\pi_{\text{oracle-1}}$	no		yes		1	one-step oracle upper bound
$\pi_{\text{oracle-look}}$	no		yes		H	cumulative-RRI headroom estimate
π_Q	yes		no		H	learned recovery over myopic scoring when headroom is positive

Table 20: Leakage-aware policy comparison. Report $J_e^{(H)}$, $G_t^{(H)}$, scene RRI, cost, invalidity, runtime, and coverage for each row.

Policy comparisons are paired by root snippet, target, candidate seed, candidate budget, and horizon. Report mean, median, bootstrap confidence intervals, and per-scene win rates for $J_e^{(H)}$, $G_t^{(H)}$, and invalidity; scene-level failures are reported separately from global averages.

4.4 Failure Interpretation

The failure interpretation is part of the research contract. If geometry or oracle labels fail, the contribution becomes a validation and one-step-scoring study. If target matching is sparse or ambiguous, target-specific RRI is reported only on validated subsets with unmatched counts and acceptance filters. If Δ_{look} is near zero, the thesis reports no measurable non-myopic headroom for the evaluated split, target set, horizon, branch factor, and candidate distribution. The next diagnosis is target matching, candidate support, and supervision scale; added model complexity is justified only after those evidence gaps are ruled out. Scaling and online-discrete tests remain interpretable only if they preserve target-specific RRI supervision.

Validation TODO: Replace planned evidence gates with actual result tables, figures, and failure cases after M1-M5 runs are complete.

Source: proposal objectives/evaluation; advisor handout

Gate: final experiments

5 Conclusion

The expected final contribution is a reproducible target-aware finite-candidate view-selection study, not a broad claim that continuous reinforcement learning has been solved for egocentric reconstruction. The thesis tests a single planning hypothesis: after oracle target-task sampling and target-specific supervision are defined, finite-candidate oracle lookahead should expose whether non-myopic target-specific RRI headroom exists, and a learned finite-horizon value model should recover part of that headroom under matched oracle re-evaluation.

The scope limit is central to the conclusion. The thesis does not claim full continuous-control NBV, online RL, real-device deployment, or replacement of target-specific point-mesh RRI by coverage, uncertainty, or semantic proxy objectives. Online discrete interaction, continuous target-then-pose policies, simulator-backed actor-critic, SceneScript, VLM planning, sparse/point backbones, and 3DGS control are escalation studies; they enter only if they preserve the finite-candidate target-specific RRI comparison.

If Δ_{look} is positive and Q_H recovers part of it under oracle-rescored selected trajectories, the thesis supports the claim that target-conditioned finite-candidate planning contains learnable non-myopic structure in the evaluated ASE/EVL regime. If headroom is near zero, the thesis instead reports a negative planning result for the evaluated candidate generator, horizon, target set, and split. If headroom exists but Q_H does not recover it, the failure analysis separates target observability, candidate support, rollout coverage, reward definition, and model-capacity explanations.

Research TODO: Rewrite the conclusion after final evidence is known. Preserve the conditional success/failure structure rather than forcing a positive planning claim.

Source: proposal risk control; advisor handout

Gate: final writing freeze

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Appendix A: Supplementary Material

Implementation TODO: Add supplementary derivations, camera-convention details, full seminar-substrate adaptation notes, generated manifest tables, and implementation-flow figures.

Source: thesis peer-review pass; seminar paper oracle/CORAL/offline-cache sections

Gate: final appendix pass

6 Draft Intake and Open Work

Archived source note: This chapter preserves substantive planning material from the retired proposal and advisor handout. It is part of the thesis seed, not final thesis prose. Each item should either graduate into the main chapters, move to an appendix, or be removed after its source decision is resolved.

Source: archived proposal and advisor handout

6.1 Schedule and Risk Control

The planned thesis window runs from 29 April 2026 to 30 September 2026. The roadmap owns the detailed Gantt chart; this draft keeps milestone exit conditions for traceability.

Dates		Milestone	Exit condition
2026-04-29	to	M0 proposal contract	Proposal, roadmap, research questions, and source policy state the same finite-candidate thesis claim.
2026-05-10			
2026-05-11	to	M1 data/oracle	Offline-store, split, pose/frame, depth/backprojection, invalidity, Rerun, and throughput checks pass before target scale-up.
2026-05-31			
2026-06-01	to	M2 one-step baseline	Scene-level VIN baseline, calibration plots, LRZ sharding plan, and Zarr rollout/Q schema are ready.
2026-06-21			
2026-06-22	to	M3 target oracle	Target RRI, V1 OBS-SEL / PRED-Q / GT-EVAL, and observed-only target selection are trusted on a small subset.
2026-07-12			
2026-07-13	to	M4 target scorer	Target-conditioned one-step scoring is compared with scene-level scoring and oracle target labels.
2026-08-09			
2026-08-10	to	M5 headroom/ Q_H	Oracle lookahead headroom is measured; Q_H is trained and oracle-evaluated if the headroom is positive.
2026-08-30			
2026-08-31	to	M6 follow-up design	Online discrete Q_H , IQL, actor-critic, hierarchy, and simulator paths are written as follow-up designs or post-M5 ablations.
2026-09-13			
2026-09-14	to	M7 experiments/writing	Final tables, figures, failure cases, coverage report, and thesis narrative are frozen.
2026-09-27			
2026-09-28	to	M8 release	Configs, smoke checks, demo path, and final PDF artifacts are reproducible.
2026-09-30			

Table 21: Retained milestone exit criteria from the proposal.

Conflict: The absolute dates are retained from the proposal source but must be checked against the current roadmap before they appear as final thesis text.

Source: proposal schedule; current roadmap

Gate: roadmap synchronization

6.2 Preliminary Thesis Outline

Chapter	Purpose	Expected evidence
Introduction	Motivate target-conditioned, quality-driven NBV for egocentric indoor reconstruction and state the finite-candidate thesis claim.	Research questions, contribution boundary, and source-backed scope.
Background	Review active perception, NBV, RRI, ASE/Project Aria, EFM3D/EVL, target-aware 3DGS, and offline value learning.	Literature synthesis with adoption/rejection decisions.
Data and geometry contracts	Describe snippets, calibration, frames, offline stores, candidates, rendered depths, backprojection, masks, and Rerun inspection.	Geometry contract report, visual diagnostics, throughput, and known limitations.
Oracle RRI and target RRI	Define scene/target distances, crop matching, invalidity, and label generation.	Label distributions, target crops, target/scene divergence, and failure cases.
Target-conditioned scoring	Present target-task encoding, candidate features, VIN-style model, ordinal/regression losses, and calibration.	Held-out ranking, top- k oracle hit, ablations, calibration, and target-specific failures.
Bounded rollout and Q_H	Compare random-valid, one-step greedy, learned one-step scorer, oracle lookahead, temperature-softmax traces, and candidate-query Q_H .	Cumulative target RRI, endpoint target gain, scene RRI, cost, invalidity, runtime, and rollout visualizations.
Discussion and conclusion	Interpret limits, scale blockers, simulator gaps, semantic/global planning, and real-device follow-up paths.	Scope-bound conclusion and reproducibility package.

Table 22: Retained preliminary chapter outline from the proposal.

Research TODO: Decide whether the final thesis keeps the current five-chapter skeleton or expands to the more detailed outline above.

Source: proposal outline; current main.typ skeleton

Gate: thesis structure freeze

6.3 Open Decisions and Questions

Open advisor decisions are the final scene-level split; symbolic target-match thresholds ($\tau_\mu, \tau_{\text{gap}}, \tau_{\text{support}}$); the CORAL-to- Q_H interface; the initial actor-visible crop descriptor ablation; the final evidence scale and subset rule; and whether any external or online scaling substrate preserves comparable mesh/oracle target-specific RRI labels. These choices affect the strength and scope of the final evidence, not the target-conditioned finite-candidate question.

Open decision: Lock final scene-level split and acceptable scale fallback.

Source: advisor handout; roadmap

Gate: M4/M5

Open decision: Lock target-match thresholds, ambiguity gap, target eligibility, and target-invalid reporting.

Source: proposal problem; advisor handout

Gate: RQ2 protocol

Open decision: Lock whether the myopic scorer freezes, slow-finetunes, or end-to-end fine-tunes when fitting residual Q_H .

Source: advisor value-model section

Gate: Q_H training plan

Open question: Can a privileged critic use GT mesh, OBB, or segmentation cues without weakening the non-privileged learned-policy result?

Source: advisor handout

Gate: critic/surrogate decision

Open question: When, if ever, do RGB, semantics, or 3DGS enter the thesis core rather than a bridge study?

Source: proposal and advisor follow-up sections

Gate: M6 scope

Implementation TODO: Validate that Rerun examples, replay integrity, shuffled-candidate evaluation, duplicate-row robustness, and valid-count sensitivity are available for final evidence reporting.

Source: advisor evidence contract

Gate: evaluation implementation audit

6.4 Future Bridges Retained from Source Material

Continuous control, external simulators, 3DGS control, SceneScript, VLM planning, IQL, CQL, BCQ, sequence decoding, soft/energy policies, PPO, SAC, privileged-teacher distillation, distributional Q_H heads, EGNN candidate graphs, sparse/point backbones, and Hestia-style target-then-pose policies remain future or post-M5 ablation material unless the finite-candidate rollout store is stable and preserves the target-specific RRI comparison.

Research TODO: Route each bridge to final discussion, appendix, or deletion after the implemented evidence is known. Do not promote these bridges to thesis-core claims without matched target-specific RRI supervision and oracle re-evaluation.

Source: proposal method; advisor adoption ledger

Gate: discussion/future-work pass

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Ehrenwoertliche Erklaerung

Ich versichere, dass ich diese Masterarbeit selbststaendig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Muenchen, TODO: Abgabedatum

Jan Duchscherer